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An introduction to Finance and to the Mathematics of arbitrage

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Part I

An introduction to corporate Finance

Chapter 1

Vocabulary

Most of the material presented in this chapter can be found in [10, 12].

1.1 Roles of financial markets

1.1.1 Generalities

1) The first main role of financial markets is to enable cash transfers between agents who aims in investing (called *investors*) and agents who are in need of funding.

   - Some examples of investors:
     * Households
     * Financial institutions
     * ...
   - Some examples of agents in need of funding:
     * Companies
     * States
     * Public administrations, local governments
     * ...

2) The second main role of financial markets is to enable risk transfers among agents on the market.

3) Finally, financial markets allows one to collect information on companies or on other financial agents.

Remark 1.1.1. *An important 4th role of financial markets is to facilitate the so-called liquidity. Liquidity is generally defined as the capacity for agents to make sell or buy orders, or to rebalance their positions.*

1.1.2 Intertemporal allocation of cash flows

At a given time, two types of agents coexist:

- Agents who aims in investing

- Agents who are in need of funding.

Cash flows transfers are physically done by emitting *securities*[1] on the market.

1.1.3 Allocation of risks

Another type of exchange on the market is the reallocation of risk in the market (which may not lead to any cash flows exchanges at the date of signing of the contract).

For instance, consider the following contract (signed/established at time $t = 0$) between two agents A and B.

- At $t = 1$, A pays an amount S to B if some random event e happened.

- At $t = 1$, B pays S_0 to A if another event $\bar{e}$ occurred.

In addition, assume that A is *rich* in case of the event e and poor if $\bar{e}$ occurs (and vice-versa for B). Hence, A carries a risk under $\bar{e}$; and B has the opposite risk under e. As a consequence, A and B have common interest to share their risks.

A classical example is the so-called *interest rate risk* which represents the exposure of an agent's positions to fluctuations of the values of one (or several) interest rate(s). Take for instance A an European exporter who has *account payables* in \$ and an European importer B having debts in \$. Consider the two random events:

- e="the exchange rate \$/€ is high (*e.g.* 1\$=1.10 €)"

- $\bar{e}$="the exchange rate \$/€ is low (*e.g.* 1\$=0.9 €)"

Hence, e is good for Agent A (and generates losses for B), whereas $\bar{e}$ is not suitable for A (but is good for B). In this situation, a contract could consists of having A to pay (resp. recieve) some amount S (resp. S_0) \$ to B at maturity time $t = 1$ if e (resp. $\bar{e}$) occurs.

[1] According to [1] "A security is a document giving title to property or claims on a financial product".

1.1.4 Efficient market Hypothesis

One of the main roles of financial markets is to provide information to agents about the financial health of companies and more generally about the whole economy. However, these information may not be deduced from the prices of different financial claims in the market. This issue is known under the name of *efficiency feature of the financial markets*.

Definition 1.1.2. *We say that a market satisfies the efficient market hypothesis if at any time all past information necessary to price contingent claims on the market are immediately observable from the prices of the financial assets of the market.*

In particular, this assumption entails that fluctuations in prices are only due to unpredictable events, which means that only agents with additional information can make abnormal profits.

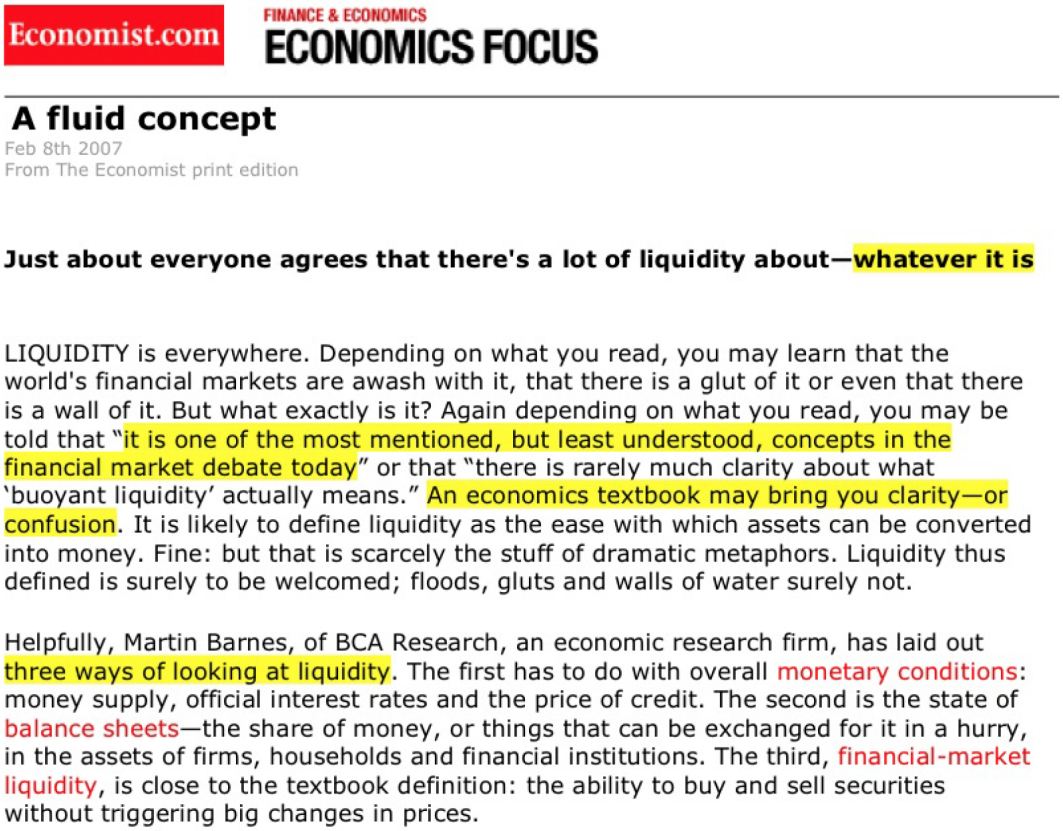
Remark 1.1.3.
- *The efficient market Hypothesis usually fails to hold in practice. Hence, Economists have developed weaker versions of this concept that go beyond this simple introduction.*

- *A complementary approach relies on the concept of rational agents which means that the market is efficient in the sense that prices reflect the future expectations of the agents according to a rational evaluation of the risks and performances of the market.*

1.1.5 Some ideas about the concept of *liquidity*

Definition 1.1.4. *The liquidity for a financial asset is the capacity for an agent to buy and/or sell shares of this asset at market price and without additional cost (transaction cost, etc...).*

In practice it is very hard to measure the liquid or illiquid feature of an asset!



Economist.com

FINANCE & ECONOMICS
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A fluid concept

Feb 8th 2007
From The Economist print edition

Just about everyone agrees that there's a lot of liquidity about—whatever it is

LIQUIDITY is everywhere. Depending on what you read, you may learn that the world's financial markets are awash with it, that there is a glut of it or even that there is a wall of it. But what exactly is it? Again depending on what you read, you may be told that "it is one of the most mentioned, but least understood, concepts in the financial market debate today" or that "there is rarely much clarity about what 'buoyant liquidity' actually means." An economics textbook may bring you clarity—or confusion. It is likely to define liquidity as the ease with which assets can be converted into money. Fine: but that is scarcely the stuff of dramatic metaphors. Liquidity thus defined is surely to be welcomed; floods, gluts and walls of water surely not.

Helpfully, Martin Barnes, of BCA Research, an economic research firm, has laid out three ways of looking at liquidity. The first has to do with overall monetary conditions: money supply, official interest rates and the price of credit. The second is the state of balance sheets—the share of money, or things that can be exchanged for it in a hurry, in the assets of firms, households and financial institutions. The third, financial-market liquidity, is close to the textbook definition: the ability to buy and sell securities without triggering big changes in prices.

1.2 Different types of financial claims/securities

1.2.1 Vocabulary

The basic object is the notion of *cash flow*.

Definition 1.2.1 (Cash flow)**.** *A cash flow is an amount of money which is transferred from an agent to another. A non-negative cash flow is referred to a cashing and a non-positive one is referred to a disbursement .*

Definition 1.2.2. *A security (or a financial claim) gives the right to its owner to a sequence of cash flows. A security is characterized by*

- *The number of cash flows.*
- *The dates at which the payments must be done. These dates can be random or deterministic.*
- *The type of the flows: we distinguish between deterministic payments and random payments. In the former case, we say that the security is riskless whereas in the latter case we say it is risky.*

- *The numéraire, that is the currency with respect to which the flow is written.*

1.2.2 Different types of securities

There exist four types of securities: *stocks (or shares of a stock)*, *bonds*, *derivatives* and *hybrid financial products.*

Shares of a stock

Definition 1.2.3. *A share of a stock is a participation to the capital of a company. It gives the right to its owner to claim dividends and to participate to the decisions made by the board of the company.*

Hence

- Stocks are risky securities.
- It generates cash flows defined by the dividends and it has a price defined by the market (mechanism of supply and demand).

Bonds

Definition 1.2.4. *A bond is a very specific form of loan. It is defined by an amount of numéraire N called the nominal (or principal or par) and by a date of reimbursement T called the maturity. In practice, the issuer of the bond receives at the subscription of the bond an issue price I and commits himself to refund the owner by paying him an amount of numéraire R at maturity. On top of that, the issuer pays at fixed pre-established times (usually every year) an interest which is called coupon according to an interest rate z which is specified at the beginning of the contract (in particular z may not be equal to the market's interest rate).*

The paradigm of a bond is when at the signature, the buyer of the bond pays I to the issuer. Then every year, at anniversary date of the signature, the issuer pays a coupon $C := N \times z$ to the buyer until maturity. At maturity, the owner pays R to the buyer. It might happen that $I = N = R$ and z is the market's rate but most of bonds do not follow this rule. In particular we say that the bond is emitted *at par* (res. *below par*/*above par*) if $I = N$ (resp. $I < N / I > N$); and that it is reimbursed *at par* (resp. *below par*/*above par*) if $R = N$ (resp. $R < N / R > N$).

One may wonder why in general it is not true that $I = N = R$. The answer is that, contrary to classical loans (between two agents), bonds are made to raise importants funds, which means that there cannot be only one investor. As a consequence, the issuer of the bond must attract investors. In addition, since the capital needed is important the maturity usually lies between 10 and 20 years. To attract investors, the issuer may propose an interest rate z higher than market's rate. Then she would for example issue the bond at

par but refund it below the par.

A fundamental example of bond is given by *zero-coupon* bonds (z.-c. bonds in short) which is one of the most important securitization and hedging tool.

Definition 1.2.5 (Zero-coupon bond (z.-c. bond))**.** *A zero-coupon bond is a bond where the interest rate of the contract z is equal to 0. Hence no coupon are paid at intermediary dates between the issue of the bond and its maturity. In that case $R = N$ and usually $N = 1$. Then an important data is the price I of the bond.*

Why are z.c. bond so important in practice?

Consider an agent who knows at time $t = 0$ that at a future date $t = T$ she will have to pay 1 unit of numéraire in \$ to a counterpart. This simple transaction is risky since buying 1\$ at maturity may be expensive depending on the value of the exchange rate €/\$ at this time. To hedge this risk, the agent can buy at time $t = 0$ a z.-c. bond (with par $N = 1$ labelled in \$) with maturity T. This costs her a certain price p \$ ($p = I$) or (equivalently some value p' labelled for instance in €). With this contract the agent has hedged her risk of paying 1\$ at a future date.

As we will see later on, the notion of z.-c. bond is related to discounting.

Hybrid financial contracts

Definition 1.2.6. *An hybrid contract is a financial contract which has characteristics of stocks and bonds. In fact most of such contracts are products which are convertible to another type of securities (for example bonds convertible into shares, or bonds that can be reimbursed using shares).*

Derivatives

Definition 1.2.7. *A derivative is a financial contract which is based on the values of one or several financial assets (which are called underlyings[2]).*

For a given derivative we define the *payoff* (of the buyer of a contract) as: the value of the underlying at Exercise time (which can be different from the maturity of the derivative) minus the cash flow at *exercise* time. As such contracts are made between two agents, if the payoff of one of the agent equals G, then the payoff of the other agent is $-G$.

Derivatives are usually divided into three categories:

- forward and futures contracts

- options

[2]In financial assets we include commodities.

- swaps

Definition 1.2.8 (Futures contract)**.** *A future is a contract between two agents in which one of them commits herself to buy (or sell) a certain amount of shares of an underlying at a given maturity T and at a given price K called the strike. A forward (or forward agreement is the same type of contract with some differences in the execution and the access to such contracts).*

Hence, the payoff for a buyer of a forward is $S_T - K$ where S_T is the value at maturity T of the underlying S. This sum may be positive or negative or 0.

Definition 1.2.9 (Option)**.** *An option is the right (and not the obligation) to buy (call) or to sell (put) a predetermined quantity of a given underlying at a date in the future (called the maturity) and at a given price (called the strike) specified at the signature of the contract.*

The classical example is a *European call option* with maturity T, on an underlying S and with *strike* K whose payoff is equal to $(S_T - K)_+ := \max\{S_T - K, 0\}$ where S_T stands for the value at maturity of the underlying. A *European put option* with maturity T, on an underlying S and with *strike* K has payoff given by $(K - S_T)_+$.

Definition 1.2.10 (Swap)**.** *A swap is an exchange of two sequences of cash flows between two agents. The contract is characterized by*

- *the duration of the swap (usually between 7 days and 10 years)*
- *the notional principal amount with respect to which the bonds are computed (and are not exchanged)*
- *the dates of exchanges (every day, week, month, etc...).*

As an example consider two agents A and B who both lend (at time $t = 0$) the same principal $N = 10\,000$ € with monthly coupons but with respect to two different interest rates: a fixed rate r for A and a floating rate r' for B (for example the 1-month Libor rate). The loan of A is riskless but with low return, whereas the investment of B is risky (as the Libor may decrease below r) but may generates higher return (when $r' > r$). Hence A and B agree on a period during which every month they will exchange the coupons they receive. Doing so, A can improve the return of her investment (with limited risk if the period is not too long) and B subscribe to a non risky situation during this period (which might help her to limit her overall exposure to risk).

We would like to point out that the type of swap we have described is called an *interest rate swap* and that other swap type contracts can be defined, as for instance *currency swap contracts*.

1.3 Different types of financial markets

Roughly speaking one usually distinguish two types of financial markets:

- Capital markets (where bonds, stocks, etc... are traded)
- Derivative markets

Note that this list is absolutely not exhaustive and that other markets exist as *commodity markets* for instance.

Another main distinction lies in the notion of *Over the Counter (OTC) markets* and *organized markets*. In an OTC market trades are done between two agents who agree on all the characteristics of the contract. On organized markets, the characteristics of the products are already pre-established.

Chapter 2

Notion of interest rate

2.1 Different ways of computing an interest

In this chapter we describe the elementary notions that define a loan. A simple way to think of the notion of interest is to interpret it as the price a consumer has to pay for not postponing his/her consumption. Hence, for an investor, interest represents the reward of the postponement of this consumption. In this section time will be measured in a discrete way. We set:

- r_t the interest rate applied on the period $[t, t+1]$ with $r_t > -1$
- I_C the initial capital (or principal) invested at time $t = 0$
- F_V the value obtained by the investment of I_C over T periods of time ($T \in \mathbb{N}$) with respect to the rates $(r_t)_{t=0,\dots,T-1}$.

There exist two modes of computing the coupons generated on each period of time.

2.1.1 Simple interest rates

The interest rate is called simple if

$$F_V = I_C \left(1 + \sum_{t=0}^{T-1} r_t\right),$$

which reduces in case where $r_t = r$, to $F_V = I_C\,(1 + rT)$. This mode of computing means that the interest generated at a given period is not added to the capital to compute the interest over the current period.

One may not be used to this system to compute the value of the interest, however it is still used nowadays for a class of banking operations over a short period of time (typically less than a year) as: trade discounts or an overdraft. For instance note that the

LIBOR[1] are quoted using the simple compounding system. Note finally, that this system is divided into two types of rates depending whether or not the interest is paid at the beginning or at the end of each period. We do not give many details here and turn to the most used way of computing interests.

2.1.2 Compound interest rates

This system is the most used and well-known. In that case at the end of any period $[t, t+1]$, the value of the interest is added to the principal for the computation of the interest period $[t, t+1]$. Hence we have

$$F_V = I_C \prod_{k=0}^{T-1} (1 + r_k),$$

which reduces to $F_V = I_C(1+r)^T$ if $r_t = r$. We refer this mechanism to compounding .

We call F_V the compounded value of I_C (with respect to $(r_k)_k$ on the period $[t, T]$).

2.2 Discounted value

In this section we consider compound interest rates.

To illustrate the notion of discounting or of discounted value we start with the simple question. Someone proposes to give you: either 100€ now (say at time $t = 0$) or 102€ at a future time (say at time $T = 3$)? To answer this question one needs to know what is the interest rate one has access to. For example assume that $r_t = r = 5\%$. Hence if one invests a principal $I_C = 100$ € at time $t = 0$ with interest rate r, over three periods of time, then he/she gets (at the end of these three periods) $100 \times 1,05^3 = 115,7625 > 102$! This elementary computation illustrates that 1€ at a given time t is not equal to 1€ at another time t' because of the interest rate. Hence the values of two cash flows at different times cannot be compared since they are not expressed in the same unit. In the previous example, we have expressed the cash flow: 100€ at $t = 0$ into the cash flow 100×1.05^3 at time $t = T$ making these two flows equivalent. One could also argue that it should make sense to determine the cash flow x€ at time $t = 0$ equivalent to 102€ at time maturity T. Rephrasing the problem, we are thus looking for the value x which, if invested over 3 periods at rate r, generates the value 102, that is the value x such that

$$x \times 1,05^3 = 102$$

[1] London Interbank Offered Rate

which obviously gives $x = 88,11144 < 100$ and then leads (hopefully) to the same conclusion. The operation of determining x is called *discounting* and then corresponds to the reciprocal operation to compounding.

Definition 2.2.1 (Discounted value of a cash flow)**.** *Let a cash flow of value F at a time t'. The discounted value V at time $t \leq t'$ of f with respect to rates $(r_k)_{k=t,\dots,t'}$ is given by:*

$$V = \frac{F}{\prod_{k=t}^{t'-1}(1+r_k)},$$

which rewrites $V = \frac{F}{(1+r_k)^{t'-t}}$ in case $r_k = r$.

Definition 2.2.2 (Discounted value of a sequence of cash flows)**.** *The discounted value of a sequence of cash flows is the sum the discounted values of each cash flow.*

2.3 Some vocabulary

We work with the compounded rates system.

Definition 2.3.1 (Periodic interest rate)**.** *We call the periodic interest rate the rate r_p applied on each period of time (daily, monthly, yearly, etc...).*

Definition 2.3.2 (Annual Percentage Rate)**.** *We call the annual percentage rate the rate r_a defined as:*

$$r_a = r_p \times k,$$

where k is the number of periods (compoundings) per year and where r_p is the periodic rate. For example, $k = 12$ if the compound unit is the month and r_p denotes the a monthly rate.

At this stage we would like to warn the reader that r_a is not, strictly speaking, an annual rate. For example assume that the period of compound is the month, then 1€ will generate after a year the value:

$$1 \times (1 + r_p)^{12}$$

which is not equal to $1 + r_a$ but to $1 + r_e$ with

$$r_e = (1 + r_p)^{12} - 1 \neq r_p \times 12.$$

r_e is called the *effective annual interest rate* and is equivalent to r_a as it generates the same capital. Then one may wonder why one considers r_a. The reason is that many operations are realized with respect to rates with annual return less than 5%. Hence doing a Taylor expansion at point 0 of the mapping $x \mapsto (1+x)^k$, one obtains: $(1+x)^k - (1 + kx) = \frac{k(k-1)}{2}x^2 + \mathrm{o}(x^2)$. Hence the difference between the interests generated by the two systems is of the order of the rate to the square which do not lead to big mistakes. This approximation was making the computations of daily bank operations much simpler in times where computation devices were not available.

Definition 2.3.3 (Effective annual interest rate)**.** *The effective annual interest rate is the rate r_e:*

$$r_e := (1 + r_p)^k - 1 = \left(1 + \frac{r_a}{k}\right)^k - 1,$$

with k the number of compoundings in one year and r_p is the periodic rate.

Definition 2.3.4 (Return of an investment)**.** *We call return of an investment of an initial capital I_C which generates the value F_V at maturity the ratio:*

$$\frac{F_V - I_C}{I_C}.$$

For instance consider the investment of an initial capital IC at a periodic rate r over one period. The final value F_V generated is equal to: $F_V = I_C(1+r)$. Hence the return equals:

$$\frac{I_C(1+r) - I_C}{I_C} = r.$$

Hence for a loan with rate r, the return is independent of the initial capital and is equal to r.

Chapter 3

A first approach to pricing: the non-random framework

3.1 A first pricing formula

We now start our analysis to derive a notion of *fair price* for financial products with these products which generate deterministic cash flows at deterministic times.

Consider a market with a benchmark bond with compounded annual rate $r \geq 0$. Let a z.c. bond with maturity T and par 1€. What should be the fair price p of this z.c. bond at time $t = 0$? For instance we can have a look at the discounted value of 1€ which simply equals to $(1 + r)^{-T}$. Assume now that $p \neq (1 + r)^{-T}$. For instance assume that $p < (1 + r)^{-T}$. In that case, an agent on the market can follow the strategy described below:

1) At $t = 0$ she borrows $(1 + r)^{-T}$ from the bank (at rate r)

2) At $t = 0$ she buys one unit of the z.c. bond (with maturity T and par 1) at price p

3) At $t = 0$ she invests the difference $(1 + r)^{-T} - p > 0$ in the riskless asset (bank) at rate r.

At maturity $t = T$, the previous actions generate the following cash flows:

1) -1 (she pays back her loan with some interest)

2) $+1$ (she receives the par from the counterpart of the z.c.)

3) $+((1 + r)^{-T} - p)(1 + r)^T = +1 - p(1 + r)^T > 0$.

Hence, assuming $p < (1+r)^{-T}$ allows an investor to generate, without initial investment and with vanishing risk, an arbitrary large benefit at maturity (just by borrowing $\alpha(1 + r)^{-T}$ and buying α units of z.c. bond at time $t = 0$ where α is any positive number). This

situation (that we will later refer to as an *arbitrage*[1]) leads to some kind of unbalance on the markets since some agents make profits with no risk and no initial capital. One can prove that the same type of opportunities can be made is $p > (1+r)^{-T}$ (in this case one sells z.-c. bonds, *i.e.* receive p at time $t = 0$ from a counterpart and will pay 1 at maturity T). As a consequence, it appears that having an interest rate r on the market imposes the values (prices) of z.c. with respect to any maturity. Hence we have proved the following:

Proposition 3.1.1. *Let a riskless bond with associated periodic rate r. Let $0 \leq t \leq T$ and $p(t,T)$ the price at time t of the z.-c.with maturity T.*

(i) If r is a compound interest rate then:

$$p(t,T) = (1+r)^{-(T-t)}.$$

(i) If r is a simple interest rate then:

$$p(t,T) = \frac{1}{1+r(T-t)}.$$

We continue our developments by considering a contract C which generates cash flows described in the table below together with two z.-c. bonds C_1 et C_2 with maturities 1 and 2 years respectively (we assume here that r is a compound interest rate):

Table 3.1: Sequence of cash flows for C, C_1 and C_2

Cash flow / Date	0	1	2
C	$-x$	$+10$	$+20$
C_1	$-(1+r)^{-1}$	$+1$	0
C_2	$-(1+r)^{-2}$	0	$+1$

As usual the main question is:

What should be the fair value x of the contract at time $t = 0$?

Following the lines of the previous example, it appears that x should be equal to the sum of the discounted values of the flows generated by C that is: $\frac{10}{1+r} + \frac{20}{(1+r)^2}$. Indeed, assume that $x > \frac{10}{1+r} + \frac{20}{(1+r)^2}$. Then one can consider the following startegy:

1) At $t = 0$ one sells C

2) At $t = 0$ one buys 10 z.-c. C_1

[1]In this section we call an arbitrage an investing strategy which does not require any initial capital at time $t = 0$ and which generates a positive capital at maturity

3) At $t = 0$ on buys 20 z.-c. C_2

4) The balance at time $t = 0$ is $+x - \frac{10}{1+r} - \frac{20}{(1+r)^2} > 0$ that is invested on the riskless account at rate r.

At time $t = 1$, one must pay 10 (because of Operation 1)), but Operation 2) brings +10 which leads to a zero profit or loss. At time $t = 2$, one pays 20 but also receive 20. In addition, Operation 4) brings $\left(+x - \frac{10}{1+r} - \frac{20}{(1+r)^2}\right)(1+r)^2 > 0$. Once again, this strategy is an arbitrage. We leave as an exercise the fact that arbitrage is allowed when $x < \frac{10}{1+r} + \frac{20}{(1+r)^2}$. Hence $x = \frac{10}{1+r} + \frac{20}{(1+r)^2}$ seems to be necessary to avoid arbitrage opportunities.

In first approximation, we have used the fact the existence of a riskless asset with interest rate r. However, we see from the formula above, that the only pricing instrument are the z.c. bonds. As a consequence we can work with the more general hypothesis where we only assume the prices of the bonds to be given by the market without any reference to an underlying rate r. This remark allows us to state the main result of this section (which is our first pricing formula).

Theorem 3.1.2 (Pricing with discounting in deterministic environment)**.** *Consider a financial market on which z.-c. bonds with par* 1 *and maturities* i *are available at (known at time* $t = 0$*) prices* C_i $i = 1, \ldots, T$*. The arbitrage price* p_A *at* $t = 0$ *of a contrat which generates a sequence of cash flows* $F_i \in \mathbb{R}$ *at dates* i*,* $i = 1, \ldots, T$ *is given by*

$$p_A := \sum_{i=1}^{T} F_i \times C_i. \tag{3.1.1}$$

In case a periodic rate r *is available on the market one has in addition that* $C_i = (1+r)^{-i}$*.*

The proof is left to the reader who is greatly encourage to practice on this simple situation.

Obviously the formula above is not completely satisfactory since it requires the dates and the values of the flow to be known at time $t = 0$. This means that we need to go further in our analysis by allowing the cash flows to be random. This objective will be addressed in the next chapters. Before going to these developments, we conclude this chapter with a re-writing of Formula (3.1.1) involving the so-called *yield curve*[2].

3.2 Yield curve

Consider a z-.c. bond with par 1, maturity T and price p_T. The return r_T of this bond is equal to:

$$r_T = \frac{1 - p_T}{p_T}.$$

[2]courbe des taux par terme

Hence we can identify this bond with a loan over the time period $[0, T]$ with periodic rate r_T. We can also compute the annual rate $z(T)$ equivalent to r_T and which is given as:

$$1 + z(T) = (1 + r_T)^{1/T},$$

leading to $z(T) := (1 + r_T)^{1/T} - 1 = p_T^{-1/T} - 1$.

Definition 3.2.1 (Yield curve)**.** *The yield curve denotes the curve of the mapping:*

$$T \mapsto z(T).$$

In case of a market where z.-c. with several maturities are available, it is possible to interpolate the points of the associated yield curve in order to approximate the price of new bonds which can then be used to hedge other products (see Exercise 3 of Exercise sheet 1).

With this notation at hand, Formula (3.1.1) rewrites as

$$p_A = \sum_{i=1}^{n} \frac{F_i}{(1 + z(t_i))^{t_i}}.$$

Part II

Mathematical analysis of Financial markets

Chapter 4

A toy model

4.1 Description of the model

We start our Mathematical analysis of financial markets with a very simple model where the value of the *risky* asset is random but follows a simple dynamic. In order to describe this model we first need to introduce the following notations.

4.1.1 Probabilistic's structure of the model

We set:

- A deterministic interest rate $r \geq 0$[1]
- A real number p with $p \in (0,1)$.
- Two real numbers d, u with $0 < d < u$.
- A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega := \{\mathfrak{T}, \mathfrak{H}\}$ ($\mathfrak{T}$ stands for "Tail" and $\mathfrak{H}$ for "Head"), $\mathcal{F} := \mathcal{P}(\Omega)$ and

$$\mathbb{P}[\{\mathfrak{T}\}] = p, \quad \mathbb{P}[\{\mathfrak{H}\}] = 1 - p,$$

  which will allow us to model the value of the risky asset.

- C a Bernouilli random variable with

$$C(\omega) := \begin{cases} u, \text{ if } \omega = \mathfrak{T}, \\ d, \text{ is } \omega = \mathfrak{H} \end{cases}, \ \omega \in \Omega.$$

With these definitions at hand, we can start to describe the financial market of interest.

[1] In reality r should be allowed to take negative values with $r > -1$, however to make our model simpler we make the assumption that r is non-negative.

4.1.2 The financial market

We consider a financial market with two assets:

- A *riskless* asset $S^0 := (1, S_1^0)$ (also called bond) defined through an interest rate r as

$$S_1^0 = 1 + r.$$

- A *risky* asset $S := (S_0, S_1)$ whose price S_1 at time $t = 1$ is given by[2]:

$$S_1 = S_0 C, \quad S_0 > 0.$$

Note that S carries out more risk than S^0 since its value at time $t = 1$ is a random variable with:

$$S_1(\omega) = \begin{cases} S_0\, u, \text{ in case } \omega = \mathfrak{T}, \\ S_0\, d, \text{ if } \omega = \mathfrak{H} \end{cases}.$$

As we assume that $u > d > 0$, the value of the price take either a "higher" level or a "lower" one compared to the initial one S_0 as shown in the figure below:

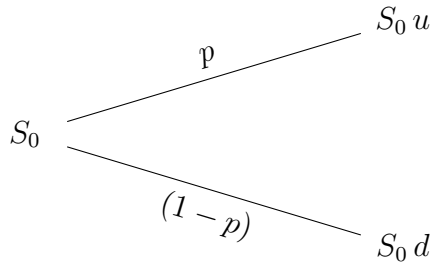


Figure 4.1: Representation of the price process S

Even though, the previous model is very simple, it will allow us to introduce the main definitions and paradigms we will develop in more complicated models and which are at the basis of Mathematical finance.

4.2 Investment strategies/portfolios

4.2.1 Some generalities

We continue the definitions with the modeling of the investment of an agent on the market described in the previous section. We would like to point out here a subtlety that might be of importance in practice, that is, we assume the agent to be "small", meaning that its action do not influence the value of the price process S.

[2] we also identify the asset with its price

Definition 4.2.1 (Portfolio: 1st definition)**.** *In our model a portfolio is a pair* $(x, \pi) \in \mathbb{R}_+ \times \mathbb{R}$ *where* x *is the initial capital of the agent and* π *denotes the number of shares of* S *she decides (at time* $t = 0$*) to hold in her portfolio.*

Hence, at $t = 0$, the investor chooses the number of shares of S she wishes to get in her portfolio for the period $[0, 1]$. To each portfolio (x, π) is associated a *wealth* process $X^{(x,\pi)}$ defined as follows.

Definition 4.2.2 (Wealth associated to a portfolio)**.** *Let* (x, π) *be a portfolio. We denote by* $X^{x,\pi} := (X_0^{x,\pi}, X_1^{x,\pi})$ *the wealth process associated to* (x, π) *and define it as:*

$$\begin{cases} X_0^{(x,\pi)} := x, \\ X_1^{(x,\pi)} := \pi S_1 + (x - \pi S_0)(1 + r) \end{cases}$$

Note that $X_1^{(x,\pi)}$ *is by definition a random variable.*

We will come back to that point later but our definition of portfolio is in fact a notion of so-called *self-financing* portfolio meaning that the numéraire (cash) which has not been invested in the risky asset S is assumed to be automatically invested in the riskless bond S^0. In other words, there are no other cash flows than those coming from S^0 and S.

4.2.2 Discounted price and wealth processes

As we have seen in Section 2.2, discounting is a fondamental notion in Finance. Here we assume that the numéraire with respect to which we discount is the riskless bond S^0. We set $\tilde{S} := (\tilde{S}_0, \tilde{S}_1)$ the discounted risky asset S with:

$$\tilde{S}_0 := S_0, \quad \tilde{S}_1 := \frac{S_1}{(1 + r)}.$$

Definition 4.2.3. *Let* (x, π) *be a portfolio. We set* $\tilde{X}^{(x,\pi)} := (\tilde{X}_0^{(x,\pi)}, \tilde{X}_1^{(x,\pi)})$ *the discounted wealth process and we have:*

$$\tilde{X}_t^{(x,\pi)} := \frac{X_t^{(x,\pi)}}{(1 + r)^t}, \quad t = 0, 1.$$

Direct computation gives that:

$$\tilde{X}_1^{(x,\pi)} = x + \pi(\tilde{S}_1 - \tilde{S}_0).$$

4.3 Arbitrage

We have seen in Chapter 3 a notion of "unbalance" (that we referred to arbitrage) on deterministic markets. In presence of randomness we need to give a more precise definition as follows.

Definition 4.3.1 (Arbitrage Opportunity)**.** *An arbitrage opportunity (in short AO) is a portfolio of the form* $(0, \pi)$ *such that:*

$$\mathbb{P}[X_1^{(0,\pi)} \geq 0] = 1 \quad \textit{and} \quad \mathbb{P}[X_1^{(0,\pi)} > 0] > 0.$$

Thus it is a strategy such that:

(i) **it does not cost anything at initial time**: $x = 0$

(ii) **it does not carry any risk** as: $\mathbb{P}[X_1^{(0,\pi)} \geq 0] = 1$

(iii) **it generates a positive profit with positive probability**: $\mathbb{P}[X_1^{(0,\pi)} > 0] > 0$.

Remark 4.3.2. *The strategy* $(0, 0)$ *is* <u>not</u> *an AO.*

Excluding AOs seems to be a necessary requirement before any analysis aiming at giving a notion of fair price for contingent claims. Hence, from now on, we would like to take the *Absence of Arbitrage Opportunity* (AAO) as a basic assumption on any model we will consider. But what is the meaning of such an assumption? That is, can we provide conditions on the parameter (p, r, u, d) defining the market so that AO are automatically excluded? We address this question in the following result.

Proposition 4.3.3. *There do not exist AOs on the market* (S^0, S) *if and only if*

$$d < 1 + r < u. \tag{4.3.1}$$

Proof. We start with the necessary condition. To this end we assume that Relation (4.3.1) is not satisfied and we will construct an AO. Saying Relation (4.3.1) fails to hold means that either $d \geq 1 + r$ or $1 + r \geq u$, which calls for studying these two cases. We treat here the case where $1 + r \geq u$ (the other case follows the same lines and is left as an exercise to the reader).

1st case: $1 + r \geq u$
Condition $1 + r \geq u$, means that the higher value of the risky asset is lower than the value of the riskless bond. This remark leads us to consider the following strategy:

$$x = 0, \quad \pi = -1,$$

whose wealth is

$$X_1^{(0,\pi)} = -S_1 + S_0(1+r) = -S_0 C + S_0(1+r) = S_0(1+r-C) \geq S_0(1+r-u) \geq 0, \ \mathbb{P}-a.s..$$

To prove that $(0, \pi)$ is an AO according to Definition 4.3.1 it just remains to prove that $\mathbb{P}[X_1^{(0,\pi)} > 0] > 0$. Indeed, we have:

$$\mathbb{P}[X_1^{(0,\pi)} > 0] = \mathbb{P}[1 + r - C > 0] \geq \mathbb{P}[C = u] = p > 0, \text{ in case where } 1 + r > u.$$

If $1 + r = u$, then

$$\mathbb{P}[X_1^{(0,\pi)} > 0] = \mathbb{P}[1 + r - C > 0] \geq \mathbb{P}[C = d] = 1 - p > 0, \text{ as } d < u.$$

Thus, $(0, \pi)$ is an AO. The reader who is interested in practicing these fondamental techniques can try to see that in fact infinitely many AO can be constructed.

2nd case: $1 + r \leq d$: left to the reader as an exercise.

We now turn to the sufficient part of the statement, that is that Condition (4.3.1) prevents the market to contain AOs. To this end, we assume that (4.3.1) holds true and that there exists an AO $(0, \pi)$ which should bring some contradiction. We have that

$$X_1^{(x,\pi)} = \pi S_1 - \pi S_0(1 + r) = S_0\pi(C - (1 + r)).$$

Assume that $\pi_1 > 0$. Then since $\mathbb{P}[X_1^{(x,\pi)} \geq 0] = 1$ it holds that $\mathbb{P}[C - (1 + r) \geq 0] = 1$ which contradicts (4.3.1). Thus $\pi \leq 0$. The same type of argument proves that $\pi < 0$ also brings a contradiction with (4.3.1). These two facts entail that the only possibility for π is $\pi = 0$. However, we have noticed in Remark 4.3.2 that $(0, 0)$ is not an AO, which concludes the proof. □

A closer look to Condition (4.3.1) enables us to give an intuitive interpretation to it. In fact, (4.3.1) just means that S is riskier than S^0 (but with possible higher return) as: Condition $u > 1 + r$ simply means that the higher value of S is bigger than the constant value of the bond (which justify to take risk in the investment), whereas Condition $d < 1 + r$ means that S may takes values which are lower than S^0 which means that it is risky.

Throughout this chapter we assume that Condition (4.3.1) **is in force.**

We now come back to the initial motivation of our analysis, that is, the pricing of financial claims in this market. By pricing we mean the process of determining a *fair price* to a given financial claim. We will see that the notion of fair price we define is the one of an *arbitrage price*.

Consider a contract between a client and a financial agent whose payoff at time $t = 1$ (say for the client) is modeled by a random variable h which depends on the value S_1. A typical example is a *call option* on S with *strike* K. An important question is then to determine the *fair price* ρ that the client should give to the agent at time $t = 0$ so that she receives the payoff h at *maturity* $t = 1$. To answer this question let assume that there exists a portfolio $(x, \bar{\pi})$ such that:

$$X_1^{(\bar{x},\bar{\pi})} = h, \quad \mathbb{P} - a.s.. \tag{4.3.2}$$

We say that h can be *replicated* by the portfolio (x, π). Note that it is not clear at first glance that such a pair $(\bar{x}, \pi)$ exists, however we will address this question later. To be able

to explain in which sense $\bar{x}$ will be a satisfying notion of price let us extend the market we have considered so far. That is we assume that an agent on the market has access to the product $\mathcal{O} := (\rho, h)$ in addition to the assets (S^0, S). This means that a portfolio now is a triple $(x, \pi, \hat{\pi})$ where $\hat{\pi}$ is the number of contracts $\mathcal{O}$ she holds in her portfolio. More precisely, having a real number $\hat{\pi}$ of $\mathcal{O}$ in a portfolio means that at initial time the agent "pays" $\hat{\pi}\rho$ and "receive" $\hat{\pi}h$ at time $t = 1$. Note that since $\hat{\pi}$ is not assumed to be non-negative, paying $\hat{\pi}\rho$ might be a positive cash flow. To make things as clear as possible, let us stress that ρ is the price indicated in the market of the contract. For a portfolio $(x, \pi, \hat{\pi})$ on the extended market $(S^0, S, \mathcal{O})$ we denote by $X_1^{(x,\pi,\hat{\pi})}$ its associated wealth at time $t = 1$ with:

$$\hat{X}_1^{(x,\pi,\hat{\pi})} := \pi S_1 + \hat{\pi}h + (x - \pi S_0 - \hat{\pi}\rho)(1 + r).$$

As before we have that $\hat{X}_0^{(x,\pi,\hat{\pi})} := x$. With these notations, the notion of AO directly applies to this market according to Definition 5.4.1. The following result is one of the main result of this section.

Theorem 4.3.4. *There are no AO on the market $(S^0, S, \mathcal{O})$ if and only if $\rho = \bar{x}$ (where $\bar{x}$ is given by (4.3.2)).*

Proof. We start with the necessary condition. Assume that $\rho > \bar{x}$ and let $\pi = +\bar{\pi}$ and $\hat{\pi} := -1$ (with $\bar{\pi}$ coming from (4.3.2)). We have:

$$\begin{aligned}
\hat{X}_1^{(0,+\bar{\pi},-1)} &= \bar{\pi}S_1 - h - \bar{\pi}S_0(1 + r) + \rho(1 + r) \\
&= \underbrace{\bar{\pi}S_1 + (\bar{x} - \bar{\pi}S_0)(1 + r)}_{=h,\ \mathbb{P}-a.s.} - h + (\rho - \bar{x})(1 + r) \\
&= (\rho - \bar{x})(1 + r) > 0, \quad \mathbb{P} - a.s..
\end{aligned}$$

So $(0, +\bar{\pi}, -1)$ is an AO which contradicts the assumption of AAO. So $\rho \leq \bar{x}$. Assume now that $\rho < \bar{x}$. One can consider the following strategy: $(0, -\bar{\pi}, +1)$ with associated wealth:

$$\begin{aligned}
\hat{X}_1^{(0,-\bar{\pi},+1)} &= -\bar{\pi}S_1 + h + \bar{\pi}S_0(1 + r) - \rho(1 + r) \\
&= \underbrace{-(\bar{\pi}S_1 + (\bar{x} - \bar{\pi}S_0)(1 + r))}_{=-h,\ \mathbb{P}-a.s.} + h + (\bar{x} - \rho)(1 + r) \\
&= (\bar{x} - \rho)(1 + r) > 0, \quad \mathbb{P} - a.s..
\end{aligned}$$

Which then gives an AO. Hence $\bar{x} = \rho$.

We turn to the sufficient condition. To this end we assume that $\bar{x} = \rho$ and that there exists an AO $(0, \pi, \hat{\pi})$, and we try to end up with a contradiction. We then have that:

$$\begin{aligned}
\hat{X}_1^{(0,\pi,\hat{\pi})} &= \pi S_1 + \hat{\pi}h - (\pi S_0 + \hat{\pi}\bar{x})(1 + r) \\
&= \pi S_1 + \hat{\pi}(\bar{\pi}S_1 + (\bar{x} - \bar{\pi}S_0)(1 + r)) - (\pi S_0 + \hat{\pi}\bar{x})(1 + r) \\
&= (\pi + \hat{\pi}\bar{\pi})S_1 - (\pi + \hat{\pi}\bar{\pi})S_0(1 + r)
\end{aligned}$$

$$= X_1^{(0,\pi+\hat{\pi}\bar{\pi})}.$$

So, $(0, \pi + \hat{\pi}\bar{\pi})$ is an AO in the market (S^0, S^1). However this is impossible by Proposition 4.3.3 which concludes the proof. □

This means that in a market free of arbitrage, replicating a payoff allows us to give a notion of fair price. In fact it is a notion of arbitrage price meaning that it is the only price such that if one adds the contract at this price in the market then the market stays arbitrage free. However, the key assumption is the replication hypothesis, and one may wonder if such an assumption is reasonable. To address this question, we fix $h : \Omega \to \mathbb{R}$ a random variable which models the payoff of the contract at time $t = 1$. Hence, there exists two real numbers h^+ et h^- such that $h(\mathfrak{T}) = h^+$ et $h(\mathfrak{H}) = h^-$. Replicating h means there exists a portfolio $(\bar{x}, \bar{\pi})$ such that $X_1^{(\bar{x},\bar{\pi})} = h$, $\mathbb{P}$-a.s. which is equivalent to:

$$(S) \begin{cases} X_1^{(\bar{x},\bar{\pi})}(\mathfrak{T}) = h(\mathfrak{T}) \\ X_1^{(\bar{x},\bar{\pi})}(\mathfrak{H}) = h(\mathfrak{H}), \end{cases}$$

which by definition of $X_1^{(\bar{x},\bar{\pi})}$ rewrites as

$$(S) \Leftrightarrow \begin{cases} \bar{\pi} S_0 u + (\bar{x} - \bar{\pi} S_0)(1 + r) = h^+ \\ \bar{\pi} S_0 d + (\bar{x} - \bar{\pi} S_0)(1 + r) = h^-. \end{cases}$$

Substracting the second line to the first one we get that

$$\bar{\pi} = \frac{h^+ - h^-}{S_0(u - d)} = \frac{h(\mathfrak{T}) - h(\mathfrak{H})}{S_1(\mathfrak{T}) - S_1(\mathfrak{H})}.$$

Plugging this expression of π in the first equation gives

$$\bar{x} = \frac{h^+}{1 + r} \frac{1 + r - d}{u - d} + \frac{h^-}{1 + r} \frac{u - (1 + r)}{u - d}.$$

As a consequence we just have proved the following result

Theorem 4.3.5. *Let h be a random variable. There exists a unique portfolio $(\bar{x}, \pi)$ such that:*

$$X_1^{(\bar{x},\bar{\pi})} = h, \ \mathbb{P} - a.s..$$

In addition:

$$\bar{x} = \frac{h(\mathfrak{T})}{1 + r} \frac{1 + r - d}{u - d} + \frac{h(\mathfrak{H})}{1 + r} \frac{u - (1 + r)}{u - d}, \quad \bar{\pi} = \frac{h(\mathfrak{T}) - h(\mathfrak{H})}{S_1(\mathfrak{T}) - S_1(\mathfrak{H})}. \tag{4.3.3}$$

We can even go a little bit further in our analysis. Let $p^* := \frac{1+r-d}{u-d}$. With this notation, one can rewrite Relation (4.3.3) as:

$$\bar{x} = \frac{h(\mathfrak{T})}{1 + r} p^* + \frac{h(\mathfrak{H})}{1 + r}(1 - p^*),$$

and that Condition (4.3.1) together with $d < u$ show that $p^* \in (0,1)$. Hence, $\bar{x}$ is equal to the expectation with respect to a probability measure $\mathbb{P}^*$ equivalent to $\mathbb{P}$ in which p is replaced by p^*. Hence, under $\mathbb{P}^*$, S_1 is still a Bernouilli random variable, so that

$$\bar{x} = \mathbb{E}^{\mathbb{P}^*}\left[\frac{h}{1+r}\right],$$

which generalizes Relation (3.1.1). This paradigm in this very simple market summarizes the main results in Mathematical Finance. Note that the existence of Measure $\mathbb{P}^*$ that we will call *Equivalent Martingale Measure* or *Risk Neutral Measure* is equivalent to the fact that p^* lies in $(0,1)$, fact which is itself equivalent to the non-arbitrage Condition (4.3.1) (under the basic assumption that $d < u$). Before closing this chapter, we would like to comment on the name *Risk Neutral Measure*. To see this terminology, we compute the expected return of S under $\mathbb{P}^*$, that is:

$$\mathbb{E}^{\mathbb{P}^*}\left[\frac{S_1 - S_0}{S_0}\right] = \frac{\mathbb{E}^{\mathbb{P}^*}[S_1] - S_0}{S_0} = \frac{S_0(up^* + d(1-p^*)) - S_0}{S_0} = r = \frac{S_1^0 - S_0^0}{S_0^0}.$$

In other words, $\mathbb{P}^*$ is a measure with respect to which the expected return of S is equal to the deterministic return of the bond which is r. All the findings of this chapter can be summarized in the following result:

Theorem 4.3.6. *The following assertions are equivalent*

(i) The market is arbitrage free

(ii) There exists a unique probability measure $\mathbb{P}^*$ equivalent to $\mathbb{P}$ such that $\mathbb{E}^{\mathbb{P}^*}\left[\frac{S_1}{1+r}\right] = S_0$

(iii) $d < 1 + r < u$,

In that case $\mathbb{P}^*(\mathfrak{T}) := p^* = \frac{1+r-d}{u-d}$.

Under one of the previous conditions, the arbitrage price $\bar{x}$ of a contract whose payoff at time $t = 1$ is modeled thanks to a random variable h is given by:

$$\bar{x} = \mathbb{E}^{\mathbb{P}^*}\left[\frac{h}{1+r}\right] = \frac{h(\mathfrak{T})}{1+r}\, p^* + \frac{h(\mathfrak{H})}{1+r}\,(1-p^*).$$

Finally, the replicating strategy or hedging strategy π of the contract is given by:

$$\bar{\pi} = \frac{h(\mathfrak{T}) - h(\mathfrak{H})}{S_1(\mathfrak{T}) - S_1(\mathfrak{H})}.$$

Application exercise: Any probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F})$ equivalent to $\mathbb{P}$ is completely determined by a real number $a \in (0,1)$ with $\tilde{\mathbb{P}}[\mathfrak{T}] := a$. Show that $\mathbb{P}^*$ is the unique martingale measure on the market presented in this chapter.

Chapter 5

Discrete time market models

5.1 Probability structure

In this course we work with a finite probability space $(\Omega, \mathcal{F}, \mathbb{P})$, that is Ω is supposed to be a finite set equipped with the σ-field $\mathcal{F} := \mathcal{P}(\Omega)$ given as the set of subsets of Ω (this σ-field is the one generated by the family $\{\{\omega\},\ \omega \in \Omega\}$) and $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$ such that $\mathbb{P}[\{\omega\}] > 0$, for any ω in Ω.

Definition 5.1.1. *A probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ is said to be equivalent to $\mathbb{P}$ if and only is $\mathbb{Q}[\{\omega\}] > 0$ for any ω in Ω.*

Warning: we warn the reader that as Ω is assumed to be finite, all the random variables defined on this space have finite moments of any order ($\mathbb{E}[|X|^p] < +\infty$ for any p, they are in fact bounded) but for more general spaces, integrability issues are very important and must be taken into consideration with a lot of care.

5.2 Discrete time financial markets

A financial market will be defined by:

- A natural number $T \geq 1$ giving the number of time periods during which investors will have access to the market (so that the financial market exists between dates $t = 0$ and $t = T$)
- A non-negative real number r which will be the interest rate of a riskless bond which will be the numéraire on the market.

A financial market is then determined by:

- A *riskless* asset $S^0 := (S^0_t)_{t=1,\dots,T}$ associated to an interest rate r with

$$S^0_t = S^0_{t-1}(1+r), \quad t = 1, \dots, T, \quad S^0_0 := 1,$$

  so that

$$S^0_t = (1+r)^t, \quad t = 1, \dots, T.$$

- $n \geq 1$ *risky* assets $S := (S^1_t, \dots, S^n_t)_{t=1,\dots,T}$, where S^i_t is a positive random variable on $(\Omega, \mathcal{F})$ which represents the price of the ith asset at a give date t ($i \in \{1, \dots, n\}$). We assume in addition, that each price S^i ($i \in \{0, \dots, n\}$) is constant on every time interval $[t, t+1[$. This simply means that quotations are made at the deterministic dates i.

A fondamental notion on financial market is the one of information. Mathematically, it is contained in the concept of *filtration* (that one could not see in the too simple one period model of Chapter 4).

Definition 5.2.1 (Filtration)**.** *A filtration is a family $\mathbb{F} := (\mathcal{F}_t)_{t \in [\![0,T]\!]}$ of sub-σ-fields of $\mathcal{F}$ such that*

$$\mathcal{F}_s \subset \mathcal{F}_t, \quad \forall (s,t) \in [\![0,T]\!]^2, \ t.q. \ s \leq t.$$

Note that the fact this family is increasing (with respect to the inclusion) is coherent with the interpretation of information.

Assumption 5.2.1. *There exists a filtration $\mathbb{F} := (\mathcal{F}_t)_{t \in [\![0,T]\!]}$ on $(\Omega, \mathcal{F})$. To make things as simple as possible, we additionally assume that $\mathcal{F}_0 := \{\emptyset, \Omega\}$ and that $\mathcal{F}_T := \mathcal{F}$.*

The assets $S^1, \dots, S^n$ are risky as their value at a positive time is random, more precisely, we assume that:

Assumption 5.2.2.

$$S^i_t \text{ is } \mathcal{F}_t\text{-measurable}, \quad \forall t \in [\![0,T]\!], \ \forall i \in \{1, \dots, n\}.$$

Before going further, we would like to give a quantum of intuition about this object and why it is useful. Assume that at a given date t, an investor is aiming to modify its portfolio for the time period $[t, t+1)$. In doing so, she will rely on the information she has so far, that is, on the values of the prices of the risky assets on $[0, t]$. Mathematically, the number of assets she will choose to hold in her portfolio will be a random variable measurable with respect to the past information which will be represented by $\mathcal{F}_t$. A classical situation is when $\mathcal{F}_t = \sigma(S^1_s, \dots, S^n_s, \ s \in \{0, \dots, t\})$.

Application exercise: Prove that by definition of a filtration, Assumption 5.2.2 entails that $\sigma(S^1_s, \dots, S^n_s, \ s \in \{0, \dots, t\}) \subset \mathcal{F}_t$. This means that $\mathcal{F}_t$ may contain strictly more information that the past values of the prices such as some exogenous information.

5.3 Investment strategy/portfolio

5.3.1 Generalities

We now define what is a portfolio or equivalently an investment strategy. Note that we assume here that the investor is of small type, meaning that her actions do not influence the values of the prices. We start with some notations.

Definition 5.3.1 (Stochastic process)**.** *A (discrete-time) stochastic process with values in* $\mathbb{R}^n$ *is an element* $\pi := (\pi_1, \cdots, \pi_T)$ *where each* $\pi_t := (\pi_t^{(1)}, \ldots, \pi_t^{(n)})$ *is a random vector valued in* $\mathbb{R}^n$ *(i.e.* $\pi_t^{(j)}$ *is a* $\mathcal{F}$*-measurable r.v. for any* $(t, j) \in \{1, \ldots, T\} \times \{1, \ldots, n\}$*).*

Definition 5.3.2 (Adapted stochastic process)**.** *A stochastic process* $\pi := (\pi_1, \cdots, \pi_T)$ *is adapted with respect to the filtration* $\mathbb{F}$ *(or simply adapted if there is no confusion about the filtration) if any r.v.* $\pi_t^{(j)}$ *(*$j \in \{1, \ldots, n\}$*) (using the notations of the previous definition) is* $\mathcal{F}_t$*-measurable for any date* t *in* $[\![0, T]\!]$*.*

Definition 5.3.3 (Predictable stochastic process)**.** *A stochastic process* $\pi := (\pi_1, \cdots, \pi_T)$ *is predictable with respect to the filtration* $\mathbb{F}$ *(or simply adapted if there is no confusion about the filtration) if any r.v.* $\pi_t^{(j)}$ *(*$j \in \{1, \ldots, n\}$*) (using the notations of the previous definition) is* $\mathcal{F}_{t-1}$*-measurable for any date* t *in* $[\![0, T]\!]$*.*

Definition 5.3.4 (Portfolio/investment strategy)**.** *A portfolio is a pair* (x, π) *where* $x \in \mathbb{R}_+$ *is called the initial capital and* π *is a* ***predictable*** *stochastic process where* $\pi_t^{(j)}$ *(*$j \in \{1, \ldots, n\}$*) stands for the number of shares of risky asset* S^j *hold in the portfolio at time* $t-1$ *(on the time period* $[t-1, t)$*).*

With these definitions at hand, at $t = 0$ the investor chooses the number of shares π_1^j of risky asset S^j she wishes to hold in her portfolio during the period $[0, 1)$ for all the assets S^j. At $t = 1$, she modifies her portfolio by choosing the quantity π_2^j of S^j in her portfolio for the period $[1, 2)$, **according to the information she has access to at this time namely the values** $S_1^{(1)}, \ldots, S_1^{(n)}$ **of** S **at time** $t = 1$**, making** π_2^j **a** $\mathcal{F}_1$**-measurable r.v.**, and so-on until time $T-1$. It is clear that notations become very quickly cumbersome, hence to simplify things we will say that π_i is the number of shares of risky asset in the portfolio. We will in fact use notations that we will allow us to make simple computations as if we were in the $n = 1$-dimensional case.

We set

$$\pi_i \cdot S_k := \sum_{j=1}^{n} \pi_i^{(j)} S_k^j, \quad (i, k) \in [\![0, T]\!]^2.$$

To each portfolio (x, π) we associate its wealth stochastic process $X^{(x,\pi)}$ as follows.

Definition 5.3.5 (Wealth associated to a portfolio)**.** *To each portfolio (x, π) we associate its wealth stochastic process $X^{(x,\pi)}$ defined as $X^{x,\pi} := (X_0^{x,\pi}, X_1^{x,\pi}, \cdots, X_T^{x,\pi})$ with:*

$$\begin{cases} X_0^{(x,\pi)} := x, \\ X_1^{(x,\pi)} := \pi_1 \cdot S_1 + (x - \pi_1 \cdot S_0)(1+r) \\ X_2^{(x,\pi)} := \pi_2 \cdot S_2 + (X_1^{(x,\pi)} - \pi_2 \cdot S_1)(1+r) \\ \cdots \\ X_t^{(x,\pi)} := \pi_t \cdot S_t + (X_{t-1}^{(x,\pi)} - \pi_t \cdot S_{t-1})(1+r), \quad t \in [\![0, T]\!] \\ \cdots \\ X_T^{(x,\pi)} := \pi_T \cdot S_T + (X_{T-1}^{(x,\pi)} - \pi_T \cdot S_{T-1})(1+r). \end{cases} \tag{5.3.1}$$

Note that by definition, at any time t, $X_t^{(x,\pi)}$ is a $\mathcal{F}_t$-measurable r.v., in other words $X^{(x,\pi)}$ is an adapted stochastic process.

From the definition of a wealth process we see that no incoming or outgoing cash flows are allowed. In other words, the agent is only allowed to share its wealth between the assets S and S^0. This is known under the name of *self-financing* portfolio. As a consequence we have:

- At $t = 0$, the agent starts the investment of her capital initial $x \geq 0$ by choosing π_1 shares of risky assets in her portfolio. This costs her (note it might be a profit if $\pi_1^{(j)} < 0$) $\pi_1 \cdot S_0$. What remains has to be invested in the riskless bond, *i.e.* the quantity $x - \pi_1 \cdot S_0$ of numéraire.

- At $t = 1$, values of S^0 and S have been updated, and so the agent computes the value of the wealth which is equal to

$$X_1^{(x,\pi)} = \underbrace{\pi_1 \cdot S_1}_{\text{part invested in } S} + \underbrace{(x - \pi_1 \cdot S_0)(1+r)}_{\text{part invested in } S^0}.$$

  Note that $x - \pi_1 \cdot S_0$ might be non-positive which means that we subscribed to a credit to buy π_1 shares of risky asset. This debt naturally comes in the value of the wealth with an interest with respect to r. Hence, r is both a loan and a compound rate. At $t = 1$, the agent also chooses the value of π_2.

- At $t = 2$, the values of S^0 and S have been updated and the wealth of the agent is equal to

$$X_2^{(x,\pi)} = \underbrace{\pi_2 \cdot S_2}_{\text{part invested in } S} + \underbrace{(X_1^{(x,\pi)} - \pi_2 \cdot S_1)(1+r)}_{\text{part invested in } S^0}.$$

  The agent chooses the value of π_3, etc...

As eluded to in the previous paragraph, the values of the π can be negative which corresponds to a *short sale*. In other words, the agent borrows to a counterpart a number of shares of S and refunds him at $t + 1$ at the price of the market.

We can re-write the expression of the wealth process which makes more clear its self-financing feature.

Proposition 5.3.6. *Let (x, π) be a portfolio. We have:*

$$X_t^{(x,\pi)} = x + \sum_{i=1}^{t} \pi_i \cdot (S_i - S_{i-1}) + \pi_i^0 (S_i^0 - S_{i-1}^0), \quad t \in [\![0, T]\!], \tag{5.3.2}$$

where $\pi_i^0 := \frac{X_{i-1}^{(x,\pi)} - \pi_i \cdot S_{i-1}}{(1+r)^{i-1}}$ denotes the number of shares invested in the riskless bond S^0 at time i.

Proof. The proof is done using induction Principle. The conclusion is straightforward when $t = 1$. Assume that the result is true at a date t with $t \geq 1$. We have:

$$\begin{aligned}
X_{t+1}^{(x,\pi)} &= \pi_{t+1} \cdot S_{t+1} + (X_t^{(x,\pi)} - \pi_{t+1} \cdot S_t)(1 + r) \\
&= \pi_{t+1} \cdot (S_{t+1} - S_t) + X_t^{(x,\pi)} + (X_t^{(x,\pi)} - \pi_{t+1} \cdot S_t) r \\
&= \pi_{t+1} \cdot (S_{t+1} - S_t) + x + \sum_{i=1}^{t} (\pi_i \cdot (S_i - S_{i-1}) + \pi_i^0 (S_i^0 - S_{i-1}^0)) + \underbrace{(X_t^{(x,\pi)} - \pi_{t+1} \cdot S_t)}_{= \pi_{t+1}^0 (1+r)^t} r \\
&= x + \sum_{i=1}^{t+1} \pi_i \cdot (S_i - S_{i-1}) + \pi_i^0 (S_i^0 - S_{i-1}^0).
\end{aligned}$$

Hence, by induction Principle the result is true for any t in $[\![0, T]\!]$. □

Remark 5.3.7. *Note that the definition we have just given corresponds to the notion of admissible portfolio in most of textbook in English. However, in textbook from the french's community, the additional requirement that $X_t^{(x,\pi)} \geq 0$, $\mathbb{P}$-a.s. for any t is considered in the definition of an admissible portfolio. This condition, has however no raison d'être from the mathematical point of view and corresponds more to some points of view in Economics. Since our analysis will not call for this extra condition, we decide to work without this extra condition.*

5.3.2 Discounted prices and portfolios

We discount the quantities of interest and we choose as a numéraire the riskless bond S^0 so that the discounted price processes $\tilde{S}$ are defined as:

$$\tilde{S}_t := \frac{S_t}{(1+r)^t} = \left(\frac{S_t^1}{(1+r)^t}, \ldots, \frac{S_t^n}{(1+r)^t} \right), \quad t \in [\![0, T]\!].$$

Definition 5.3.8. *Let (x, π) a portfolio. We set $\tilde{X}^{(x,\pi)} := (\tilde{X}_t^{(x,\pi)})_{t \in [\![0,T]\!]}$ the discounted wealth process associated to (x, π):*

$$\tilde{X}_t^{(x,\pi)} := \frac{X_t^{(x,\pi)}}{(1+r)^t}, \quad t \in [\![0, T]\!].$$

Proposition 5.3.9. *Let (x, π) a portfolio. We have:*

$$\tilde{X}_t^{(x,\pi)} = x + \sum_{i=1}^{t} \pi_i \cdot (\tilde{S}_i - \tilde{S}_{i-1}), \quad t \in [\![0, T]\!].$$

In other words, the discounted value of the wealth of a portfolio is its wealth where one replaces S^0 by the riskless bond $\tilde{S}^0 := 1$ (so with vanishing interest rate) and S by $\tilde{S}$.

Proof. Let (x, π) a portfolio. Once again the property is established using the induction Principle using Definition 5.3.5. For $t = 1$, one has:

$$\begin{aligned}
\tilde{X}_1^{(x,\pi)} &= \frac{X_1^{(x,\pi)}}{(1+r)} \\
&= \pi_1 \cdot \tilde{S}_1 + (x - \pi_1 \cdot S_0) \\
&= x + \pi_1 \cdot (\tilde{S}_1 - \tilde{S}_0).
\end{aligned}$$

Assume now that the property is true for $t \in \{1, \dots, T-1\}$. We have using (5.3.1) that

$$\begin{aligned}
\tilde{X}_{t+1}^{(x,\pi)} &= \frac{X_{t+1}^{(x,\pi)}}{(1+r)^{t+1}} \\
&= \pi_{t+1} \cdot \tilde{S}_{t+1} + \frac{X_t^{(x,\pi)} - \pi_{t+1} \cdot S_t}{(1+r)^t} \\
&= \pi_{t+1} \cdot (\tilde{S}_{t+1} - \tilde{S}_t) + \tilde{X}_t^{(x,\pi)} \\
&= x + \sum_{i=1}^{t+1} \pi_i \cdot (\tilde{S}_i - \tilde{S}_{i-1}),
\end{aligned}$$

using the induction hypothesis. Hence by induction Principle, the property is establishes at any time t. □

5.4 Arbitrage

As mentioned several times our objective is give a notion of *fair price* whatever it means. This concept will have to be understood in the sense of *arbitrage price*, which is the price of a contract such that if the contract together with this price are added to an arbitrage free market the latter stays free of arbitrage.

Definition 5.4.1 (Arbitrage Opportunity (AO))**.** *An Arbitrage Opportunity (AO) is a portfolio of the form $(0, \pi)$ such that:*

$$\mathbb{P}[X_T^{(0,\pi)} \geq 0] = 1 \quad \textit{et} \quad \mathbb{P}[X_T^{(0,\pi)} > 0] > 0.$$

Hence it is an investment strategy which:

(i) **has no initial cost**: $x = 0$

(ii) **is riskless** as: $\mathbb{P}[X_T^{(0,\pi)} \geq 0] = 1$

(iii) **generates a positive profit at terminal time T with positive probability**: $\mathbb{P}[X_T^{(0,\pi)} > 0] > 0$.

Remark 5.4.2. *Note that $(0, \mathbf{0}_n)$ is not an AO, with $\mathbf{0}_n := (0, \ldots, 0)^\top$, where $x^\top$ denotes the transpose of a given vector x.*

Definition 5.4.3. *A financial market is said to be viable is there is no AO on this market.*

Since the *absence of arbitrage opportunity (AOA)* seems to be a prerequisite to any effort to define a fair price concept one is interested in deriving a condition which ensures the (AOA) on a given financial market. Such a condition has been derived as (4.3.1) in the toy model of Chapter 4 which turned out to be a necessary and sufficient condition. We have also found that it was equivalent to the existence of an *equivalent martingale measure (EMM)* on the market with respect to which the discounted price processes where martingales. This characterization will, in turn, be a general necessary and sufficient condition for discrete financial markets to (AOA). This fundamental result is known under the name of *1st fundamental theorem of asset pricing* and has been established in this form in [6][1]. Before going to the fundamental theorem of asset pricing (see Theorem 5.4.5 below) we introduce the following definition.

Definition 5.4.4 (Equivalent Martingale Measure (EMM) or Risk Neutral Measure)**.** *An equivalent martingale measure EMM also called Risk Neutral Measure is a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \mathcal{F})$ equivalent to $\mathbb{P}$ in the sense of Definition 5.1.1 such that under $\tilde{\mathbb{P}}$, $\tilde{S}$ is a martingale.*

Theorem 5.4.5 (1st fundamental theorem of asset pricing)**.** *The market (S^0, S) is arbitrage free if and only if there exists an EMM.*

Proof. We start with the sufficient part of the statement. To this end we use a *reductio ad absurdum* argument, that is, we assume that the there exists an EMM $\tilde{\mathbb{P}}$ and that there exists an AO $(0, \pi)$. We first prove that the discounted wealth process $\tilde{X}^{(0,\pi)}$ is a $\tilde{\mathbb{P}}$-martingale. To this end, as we consider discrete-time processes, it is enough to prove that:

$$\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_t^{(0,\pi)} - \tilde{X}_{t-1}^{(0,\pi)} | \mathcal{F}_{t-1}] = 0, \quad \forall 1 \leq t \leq T. \tag{5.4.1}$$

Let $1 \leq t \leq T$. Proposition 5.3.9 entails that:

$$\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_t^{(0,\pi)} - \tilde{X}_{t-1}^{(0,\pi)} | \mathcal{F}_{t-1}] = \mathbb{E}^{\tilde{\mathbb{P}}}\left[\pi_t \cdot (\tilde{S}_t - \tilde{S}_{t-1}) | \mathcal{F}_{t-1}\right]$$

[1] In case where Ω is not supposed to be finite the result is due to [5, 8] where the notion of arbitrage is replaced by a stronger notion known under the name of "no free lunch". The general version with the no-arbitrage assumption has been derived in [3].

$$
\begin{aligned}
&= \mathbb{E}^{\tilde{\mathbb{P}}}\left[\sum_{i=1}^{n} \pi_t^{(i)}(\tilde{S}_t^{(i)} - \tilde{S}_{t-1}^{(i)})\Big|\mathcal{F}_{t-1}\right] \\
&= \sum_{i=1}^{n} \mathbb{E}^{\tilde{\mathbb{P}}}\left[\pi_t^{(i)}(\tilde{S}_t^{(i)} - \tilde{S}_{t-1}^{(i)})\Big|\mathcal{F}_{t-1}\right] \\
&= \sum_{i=1}^{n} \pi_t^{(i)}\mathbb{E}^{\tilde{\mathbb{P}}}\left[(\tilde{S}_t^{(i)} - \tilde{S}_{t-1}^{(i)})\Big|\mathcal{F}_{t-1}\right], \quad \text{as } \pi_t^i \text{ is } \mathcal{F}_{t-1} - \text{measurable} \\
&= \sum_{i=1}^{n} \pi_t^{(i)} \underbrace{\mathbb{E}^{\tilde{\mathbb{P}}}\left[(\tilde{S}_t^{(i)} - \tilde{S}_{t-1}^{(i)})\Big|\mathcal{F}_{t-1}\right]}_{=0}, \quad \text{as } \tilde{S}^i \text{ is a } \tilde{\mathbb{P}} - \text{martingale} \\
&= 0.
\end{aligned}
$$

Hence Relation (5.4.1) is established. Note that to derive it we did not make use of the fact that $(0, \pi)$ is an AO and hence that it holds true for any portfolio. In addition, as $(0, \pi)$ is an AO and since $\tilde{\mathbb{P}}$ and $\mathbb{P}$ are equivalent measures, we have that $\tilde{X}_T^{(0,\pi)} \geq 0$, $\tilde{\mathbb{P}}$-a.s. and $\tilde{\mathbb{P}}[\tilde{X}_T^{(0,\pi)} > 0] > 0$, which implies that $\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_T^{(0,\pi)}] > 0$. However this contradicts the martingale property of $\tilde{X}^{(0,\pi)}$ under $\tilde{\mathbb{P}}$ as it implies that:

$$\mathbb{E}^{\mathbb{P}^*}[\tilde{X}_T^{(0,\pi)}] = \tilde{X}_0^{(0,\pi)} = 0.$$

So the existence of a EMM excludes the existence of an AO.

We now turn to the necessary part of the result which is more difficult as we will have to construct an EMM $\tilde{\mathbb{P}}$. Recall that by definition, a measure $\tilde{\mathbb{P}}$ equivalent to $\mathbb{P}$ is an EMM if and only if $\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{S}_t - \tilde{S}_{t-1}|\mathcal{F}_{t-1}] = 0$ for any t. We proceed in two steps.

Step 1: construction of a measure $\tilde{\mathbb{P}}$.
The set $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is a vector space which can be equipped with the inner produit $\langle \cdot, \cdot \rangle$ with:

$$\langle X, Y \rangle := \mathbb{E}[XY], \quad \forall (X, Y) \in \left(L^2(\Omega, \mathcal{F}, \mathbb{P})\right)^2.$$

We set $\mathcal{P} := \{X \in L^2(\Omega, \mathcal{F}, \mathbb{P}),\ X \geq 0,\ \mathbb{E}[X] = 1\}$. Note that $\mathcal{P}$ is a convex set and it is compact with respect to the norm induced by the inner product. Now we consider the set of attainable wealths with zero initial capital that is:

$$\mathcal{W} := \{\tilde{X}_T^{(0,\pi)},\ \pi \text{ a predictable process}\}.$$

We have that $\mathcal{P} \cap \mathcal{W} = \emptyset$. Indeed, if the intersection is not empty, let X be an element of $\mathcal{P}$ and $\mathcal{W}$, then it is the terminal wealth of an AO which contradicts our assumption. Set $C := \mathcal{P} - \mathcal{W} := \{p - w,\ p \in \mathcal{P},\ w \in \mathcal{W}\}$. Since $\mathcal{P} \cap \mathcal{W} = \emptyset$, 0 does not belong to C. In addition, C is a closed convex set. Let $\tilde{X}$ be the projection of 0 on C. Then by definition of the projection, we have that:

$$\langle Y - \tilde{X}, -\tilde{X} \rangle \leq 0, \quad \forall Y \in C,$$

which means that for any Y in C, $\langle \tilde{X}, Y\rangle \geq \langle \tilde{X}, \tilde{X}\rangle > 0$ (as $\tilde{X} \neq 0$). As a consequence for any $Z_1 \in \mathcal{P}$, $Z_2 \in \mathcal{W}$,

$$\langle \tilde{X}, Z_1\rangle \geq \langle \tilde{X}, \tilde{X}\rangle + \langle \tilde{X}, Z_2\rangle.$$

Fixing Z_1 in $\mathcal{P}$ and replacing Z_2 by λZ_2 with λ a real number, we have that $\langle \tilde{X}, Z_2\rangle = 0$ for any Z_2 in $\mathcal{W}$. So to summarize, we have found a r.v. $\tilde{X}$ such that:

$$\mathbb{E}[\tilde{X}\tilde{X}_T^{(0,\pi)}] = 0, \quad \forall \pi \text{ portfolio}, \tag{5.4.2}$$

$$\mathbb{E}[\tilde{X}] = \langle \tilde{X}, 1\rangle \geq \langle \tilde{X}, \tilde{X}\rangle > 0, \quad \text{as } 1 \in \mathcal{P}.$$

So we set $\tilde{\mathbb{P}}$ the measure on $(\Omega, \mathcal{F})$ defined by:

$$\tilde{\mathbb{P}}[\{\omega\}] := \frac{\tilde{X}(\omega)\mathbb{P}[\{\omega\}]}{\mathbb{E}[\tilde{X}]}, \quad \forall \omega \in \Omega.$$

Step 2: $\tilde{S}$ is a $\tilde{\mathbb{P}}$-martingale.
Let t. We have to prove that $\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{S}_t - \tilde{S}_{t-1}|\mathcal{F}_{t-1}] = 0$. By definition of the conditional expectation operator, this relation follows if we prove that:

$$\mathbb{E}^{\tilde{\mathbb{P}}}[\mathbf{1}_A(\tilde{S}_t - \tilde{S}_{t-1})] = 0, \quad \forall A \in \mathcal{F}_{t-1}.$$

Let A in $\mathcal{F}_{t-1}$ and π the strategy defined as: $\pi_t := \mathbf{1}_A$ and $\pi_s = 0$ for $s \neq t$. The process π is by definition a predictable process and we have that $\tilde{X}_T^{(0,\pi)}$ is an element of $\mathcal{W}$. Hence Relation (5.4.2) implies that $\mathbb{E}^{\tilde{\mathbb{P}}}[\mathbf{1}_A(\tilde{S}_t - \tilde{S}_{t-1})] = 0$ which concludes the proof. □

5.5 Pricing and hedging of replicable European contingent claims

5.5.1 General pattern

Given a contract whose payoff at time T is given by a $\mathcal{F}_T$-measurable random variable h, we aim in giving a notion of fair price to this contract. In what follows, we will refer to h as a European contingent claim. For notational convenience we set $\mathcal{P} := \{\tilde{\mathbb{P}} \text{ EMM}\}$.

Definition 5.5.1 (European contingent claim)**.** *We call an European contingent claim, any $\mathcal{F}_T$-measurable random variable h. It corresponds to the payoff that an agents pays or received at a given time*[2]*.*

Definition 5.5.2 (Replicable contingent claim)**.** *A contingent claim h is said to be replicable on the market (S^0, S) if there exists a portfolio (x^h, π^h) such that*

$$X_T^{(x^h,\pi^h)} = h, \ \mathbb{P} - a.s..$$

A strategy π^h is called a hedging strategy.

[2]As a consequence, these products are different of American contingent claims where the exercise time is not specified in the contact.

We start with the following proposition.

Proposition 5.5.3. *Let (x, π) and (x', π') two portfolios such that:*

$$X_T^{(x,\pi)} = X_T^{(x',\pi')}, \quad \mathbb{P} - a.s..$$

Then for any time t in $[\![0, T]\!]$ it holds that:

$$X_t^{(x,\pi)} = X_t^{(x',\pi')}, \quad \mathbb{P} - a.s..$$

In particular $x = x'$.
Note that it does not mean that $\pi = \pi'$ (in general this fact is not true).

Proof. Let $\tilde{\mathbb{P}}$ an EMM. Then the processes $\tilde{X}^{(x,\pi)}$ and $\tilde{X}^{(x',\pi')}$ are $\tilde{\mathbb{P}}$-martingales with the same value at time T. Hence

$$\tilde{X}_t^{(x,\pi)} = \tilde{X}_t^{(x',\pi')}, \quad \forall t \ \mathbb{P} - a.s.$$

from which the result follows my multiplying by S_t^0 in the equality. □

As in Theorem 4.3.4 the value x^h is the arbitrage price of h, meaning that if the market (S^0, S) is arbitrage free, then the extended market $(S^0, S, (x^h, h))$ is arbitrage-free. More precisley, we have:

Theorem 5.5.4. *Let h be a replicable contingent claim with replicating portfolio (x^h, π^h). Then x^h is the arbitrage price of h and:*

$$x^h = \mathbb{E}^{\tilde{\mathbb{P}}}\left[\frac{h}{S_T^0}\right], \quad \forall\, \tilde{\mathbb{P}} \in \mathcal{P}. \tag{5.5.1}$$

More generally the arbitrage price ρ_t^h of h at a given time $t \in [\![0, T]\!]$ is given by:

$$\rho_t^h = S_t^0 \mathbb{E}^{\tilde{\mathbb{P}}}\left[\frac{h}{S_T^0}\Big|\mathcal{F}_t\right], \quad \forall\, \tilde{\mathbb{P}} \in \mathcal{P}.$$

A hedging strategy π^h is given by:

$$\pi_t^h := \mathbb{E}^{\tilde{\mathbb{P}}}\left[(\tilde{X}_t^{(x^h,\pi^h)} - \tilde{X}_{t-1}^{(x^h,\pi^h)})(\tilde{S}_t - \tilde{S}_{t-1})^\top | \mathcal{F}_{t-1}\right] P_t^{-1} \mathbf{1}_{\{P_t \text{ is positive definite}\}}, \quad t \geq 1, \tag{5.5.2}$$

where $P_t := \mathbb{E}^{\tilde{\mathbb{P}}}[(\tilde{S}_t - \tilde{S}_{t-1})(\tilde{S}_t - \tilde{S}_{t-1})^\top | \mathcal{F}_{t-1}]$ and P_t^{-1} denotes its inverse if it is positive-definite $\mathbb{P}$-a.s..

Proof. We set $\rho_t := X_t^{(x^h,\pi^h)}$. We consider the financial market: (S^0, S, ρ). By Theorem 5.4.5, this market is viable if and only if there exists an EMM measure on this market. Since the latter contains the original market, an EMM on the enlarged market has to be an EMM for the initial market. In fact we will prove that the set of EMM for the initial

market and the set of EMM for the enlarged market coincide. Indeed, let $\tilde{\mathbb{P}}$ be an EMM on the original market. $\tilde{\mathbb{P}}$ is an EMM on the enlarged market if the discounted process $\tilde{\rho}$ is a $\tilde{\mathbb{P}}$-martingale. Let $1 \leq t \leq T$. We have that:

$$\begin{aligned}\mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{\rho}_t - \tilde{\rho}_{t-1}|\mathcal{F}_{t-1}] &= \mathbb{E}^{\tilde{\mathbb{P}}}[\pi_t \Delta_t \tilde{S}|\mathcal{F}_{t-1}] \\ &= \pi_t \mathbb{E}^{\tilde{\mathbb{P}}}[\Delta_t \tilde{S}|\mathcal{F}_{t-1}] \\ &= 0,\end{aligned}$$

as $\tilde{S}$ is a $\tilde{\mathbb{P}}$-martingale. As a consequence, at any time t we have that

$$\rho_t = S_t^0 \tilde{\rho}_t = S_t^0 \mathbb{E}^{\tilde{\mathbb{P}}}\left[\frac{h}{S_T^0}\Big|\mathcal{F}_t\right].$$

Concerning the replication strategy, let $t \geq 1$. By definition,

$$\tilde{X}_t^{(x^h,\pi^h)} - \tilde{X}_{t-1}^{(x^h,\pi^h)} = \pi_t^h(\tilde{S}_t - \tilde{S}_{t-1}).$$

Multiplying this relation by $(\tilde{S}_t - \tilde{S}_{t-1})^\top$, taking the conditional expectation and recalling that π^h is predictable, gives the result. □

5.5.2 Complete markets

Definition 5.5.5. *A financial market is called a complete market if any financial claim is replicable on the market.*

The main result in this section is the following characterization of complete markets.

Theorem 5.5.6. *A viable financial market (S^0, S) is complete if and only if there exists a unique EMM.*

Proof. Since the market is viable there exists at least one EMM. We have to prove that the market is complete if and only if the set of EMM is a singleton. We start by proving that uniqueness of an EMM entails that the market is complete. To this end, assume that there exists a unique EMM $\tilde{\mathbb{P}}$ and that the market is not complete. By definition, it means that the set $\mathcal{W}$ defined as:

$$\mathcal{W} := \{X_T^{(x,\pi)},\ (x,\pi) \text{ portfolio}\}$$

is a vector space strictly included in $L^0(\Omega, \mathcal{F}_T)$ which is an Hilbert space equipped with the inner product

$$\langle X, Y\rangle := \mathbb{E}^{\tilde{\mathbb{P}}}[XY], \quad X, Y \in L^2(\Omega, \mathcal{F}_T).$$

Hence there exists a contingent claim h such that

$$h = X_T^{(x,\pi)} + Y,$$

where (x, π) is a portfolio and Y belongs to $\mathcal{W}^\top$ and $Y \neq 0$. In particular $\mathbb{E}^{\tilde{\mathbb{P}}}[XY] = 0$ for any X in $\mathcal{W}$. Define the probability measure:

$$\mathbb{P}^* := (1 + Y)\tilde{\mathbb{P}}.$$

Let $0 \leq s \leq t \leq T$ and A in $\mathcal{F}_s$. We have that:

$$\mathbb{E}^{\mathbb{P}^*}[(\tilde{S}_t - \tilde{S}_s)\mathbf{1}_A] = \mathbb{E}^{\mathbb{P}^*}[\tilde{X}_t^{(0,\tau)}]$$

with $\tau_r := \mathbf{1}_A$ for $r \in \{s+1, \cdots, t\}$ and $\tau_r = 0$ otherwise. In particular $X_t^{(0,\tau)}$ is an element of $\mathcal{W}$. Hence we have that:

$$\mathbb{E}^{\mathbb{P}^*}[(\tilde{S}_t - \tilde{S}_s)\mathbf{1}_A] = \mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_t^{(0,\tau)}] + \mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_t^{(0,\tau)}Y] = 0.$$

Hence $\mathbb{E}^{\mathbb{P}^*}[\tilde{S}_t - \tilde{S}_s|\mathcal{F}_s] = 0$ which proves that $\mathbb{P}^*$ is an EMM which contradicts the assumption.

We now prove the converse part of the result. Namely we assume that the market is complete and prove that there exists only one EMM. Let $\tilde{\mathbb{P}}$ and $\mathbb{P}^*$ two EMM and let A in $\mathcal{F}_T$. Consider the contingent claim $h := \mathbf{1}_A S_T^0$. Since the market is complete there exists a portfolio (x, π) such that:

$$h = X_T^{(x,\pi)}, \quad \mathbb{P} - a.s..$$

Hence, using the fact that $\tilde{\mathbb{P}}$ and $\mathbb{P}^*$ are EMMs we have that:

$$\tilde{\mathbb{P}}[A] = \mathbb{E}^{\tilde{\mathbb{P}}}[\tilde{X}_T^{(x,\pi)}] = x = \mathbb{E}^{\mathbb{P}^*}[\tilde{X}_T^{(x,\pi)}] = \mathbb{P}^*[A],$$

which concludes the proof. □

Chapter 6

The Cox-Ross-Rubinstein (or Binomial model)

6.1 Description of the model

In this chapter we present a particular example of discrete-time market model of Chapter 5 which extends the introductory market described in Chapter 4.

6.1.1 Probabilistic structure of the model

We set:

- An interest rate $r \geq 0$

- A natural number $T \geq 1$, so that the financial market is defined on $[\![0, T]\!]$.

- A real number p with $p \in (0, 1)$.

- Two real numbers d, u with $0 < d < u$.

- A probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega := \{\mathfrak{T}, \mathfrak{H}\}^T$, $\mathcal{F} := \mathcal{P}(\Omega)$, and

$$\mathbb{P}[\{(\omega_1, \dots \omega_T)\}] = p^{\#\{i,\, \omega_i = \mathfrak{T}\}}(1-p)^{\#\{i,\, \omega_i = \mathfrak{H}\}}, \quad \omega_i \in \{\mathfrak{T}, \mathfrak{H}\},\ i = 1, \dots, T.$$

- T independent Bernouilli random variables $C_1, \dots, C_T$ with

$$C_i((\omega_1, \dots \omega_T)) := C_i((\omega_i)) = \begin{cases} u, \text{ if } \omega_i = \mathfrak{T}, \\ d, \text{ if } \omega_i = \mathfrak{H} \end{cases}, \quad i = 1 \dots, T,\ (\omega_1, \dots \omega_T) \in \Omega.$$

- We set $\mathcal{F}_t := \sigma(C_1, \dots, C_t)$, for $t \in [\![0, T]\!]$ (note that $\mathcal{F}_T = \mathcal{F}$).

6.1.2 The market model

The financial market is composed of:

- A bond $S^0 := (S^0_t)_{t=1,\dots,T}$ with periodic compound rate r:

$$S^0_t = S^0_{t-1}(1+r), \quad t = 1, \dots, T, \quad S^0_0 := 1,$$

so

$$S^0_t = (1+r)^t, \quad t = 1, \dots, T.$$

- A risky asset $S := (S_t)_{t=1,\dots,T}$ such that

$$S_t = S_{t-1}C_t, \quad t = 1, \dots, T, \quad S_0 > 0.$$

Hence:

$$S_t((\omega_1, \dots, \omega_t)) = \begin{cases} S_{t-1}((\omega_1, \dots, \omega_{t-1}))\, u, \text{ if } \omega_t = \mathfrak{T}, \\ S_{t-1}((\omega_1, \dots, \omega_{t-1}))\, d, \text{ if } \omega_t = \mathfrak{H} \end{cases}, \quad t = 1 \dots, T.$$

As $u > d > 0$, the parameter u is the *up factor* of the price process and d the *down factor*.

Remark 6.1.1. *Note that in this model, several sample paths lead to the same value of the price at a given time as one can see in the figure below (with $T = 2$).*

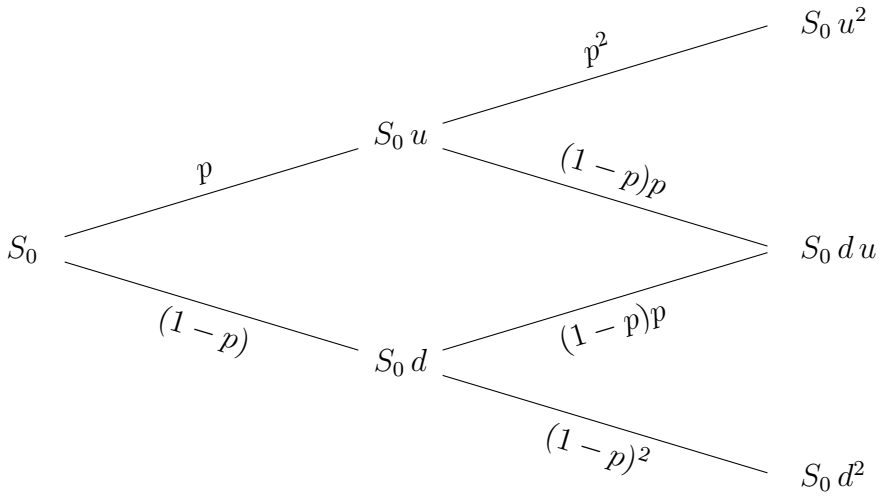


Figure 6.1: The binomial tree in case $T = 2$

Application exercise: Give the number of possible outcomes for the value of S_t for any t in $[\![0, T]\!]$.

Obviously this market model is a particular example of the general framework of Chapter 5.

Proposition 6.1.2. *The CRR market is viable if and only if*

$$d < 1 + r < u. \tag{6.1.1}$$

Proof. Assume first that the market is viable (*i.e.* there is no AO). Then by Theorem 5.4.5, there exists an EMM $\mathbb{P}^*$. Let $0 \leq t \leq T-1$. We have that:

$$\mathbb{E}^{\mathbb{P}^*}[\tilde{S}_{t+1}|\mathcal{F}_t] = \tilde{S}_t,$$

from which

$$\mathbb{E}^{\mathbb{P}^*}[C_{t+1}|\mathcal{F}_t] = 1 + r.$$

Taking expectation implies that:

$$\mathbb{E}^{\mathbb{P}^*}[C_{t+1}] = 1 + r.$$

Hence,

$$\mathbb{P}^*[C_{t+1} = u] = \frac{1 + r - d}{u - d}.$$

As $\mathbb{P}^*$ is absolutely continuous with respect to $\mathbb{P}$, we have that $\mathbb{P}^*[C_{t+1} = u] \in (0, 1)$ which leads to (6.1.1).

We now turn to the sufficient part of the result. Let:

$$p^* := \frac{1 + r - d}{u - d}.$$

Condition (6.1.1) implies that p^* belongs to $(0, 1)$. Set:

$$\mathbb{P}^*[\{(\omega_1, \ldots, \omega_T)\}] := (p^*)^{\#\{i;\omega_i = \mathfrak{T}\}}(1 - p^*)^{\#\{i;\omega_i = \mathfrak{H}\}}, \quad \forall (\omega_1, \ldots, \omega_T) \in \Omega.$$

We let as an exercise that this expression defines a probability measure. We will prove that it is an EMM. To this end, we first prove that $(C_1, \cdots, C_T)$ are independent with respect to $\mathbb{P}^*$. As Ω is finite, it is enough to prove that for any $x_j \in \{d, u\}$ ($j \in [\![0, T]\!]$) it holds that:

$$\mathbb{P}^*[\cap_{j \in [\![0,T]\!]} C_j = x_j] = \prod_{j \in [\![0,T]\!]} \mathbb{P}^*[C_j = x_j].$$

For notational purposes, we set: $\Phi(u) := \mathfrak{T}$ and $\Phi(d) := \mathfrak{H}$ so that: $\{C_j = x_j\} = \{\omega,\ \omega_j = \Phi(x_j)\}$. Hence:

$$\mathbb{P}^*[\cap_{j \in [\![0,T]\!]} C_j = x_j]$$

$$= \sum_{(\omega_1,\cdots,\omega_T)=(\Phi(x_1),\cdots,\Phi(x_d))} (p^*)^{\#\{j\ \Phi_j(x_j)=\mathfrak{T}\}}(1-p^*)^{\#\{j,\ \Phi_j(x_j)=\mathfrak{H}\}}$$
$$= \prod_{j\in[\![0,T]\!]} \mathbb{P}^*[C_j = x_j].$$

Finally, we can prove that $\tilde{S}$ is a $\mathbb{P}^*$-martingale. Let t in $[\![0,T]\!]$.

$$\begin{aligned}\mathbb{E}^{\mathbb{P}^*}[\tilde{S}_{t+1}|\mathcal{F}_t] &= \tilde{S}_t\mathbb{E}^{\mathbb{P}^*}[(1+r)^{-1}C_{t+1}|\mathcal{F}_t]\\ &= \tilde{S}_t\mathbb{E}^{\mathbb{P}^*}[(1+r)^{-1}C_{t+1}|\sigma(C_1,\cdots,C_t)]\\ &= \tilde{S}_t\mathbb{E}^{\mathbb{P}^*}[(1+r)^{-1}C_{t+1}]\\ &= \tilde{S}_t(1+r)^{-1}(up^*+d(1-p^*))\\ &= \tilde{S}_t.\end{aligned}$$

□

Proposition 6.1.3. *The CRR market is complete, and the unique EMM $\mathbb{P}^*$ is given by:*

$$\mathbb{P}[\{\omega_1,\dots\omega_T\}] = p^{\#\{i,\ \omega_i=\mathfrak{T}\}}(1-p)^{\#\{i,\ \omega_i=\mathfrak{H}\}},\quad \omega=(\omega_1,\dots\omega_T)\in\Omega,\ i=1,\dots,T,$$

with

$$p^* := \frac{1+r-d}{u-d}.$$

Proof. According to Theorem 5.5.6, we will prove that $\mathbb{P}^*$ defined above is the only EMM. Let $\mathbb{Q}$ be an EMM. We first prove that $(C_1,\cdots,C_T)$ are independent with respect to $\mathbb{Q}$. As $\tilde{S}$ is a $\mathbb{Q}$-martingale, for t in $\{0,\cdots,T_1\}$ it holds that:

$$\mathbb{E}^{\mathbb{Q}}[\tilde{S}_{t+1}|\mathcal{F}_t] = \tilde{S}_t$$

which is equivalent to

$$\mathbb{E}^{\mathbb{Q}}[C_{t+1}|\mathcal{F}_t] = 1+r. \tag{6.1.2}$$

Let $(x_1,\cdots,x_T)\in\{d,u\}^T$. Using this property we can then deduce that:

$$\begin{aligned}\mathbb{Q}\left[\cap_{j\in[\![0,T]\!]}C_j = x_j\right] &= \mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{C_1=x_1}\times\ldots\times\mathbf{1}_{C_T=x_T}\right]\\ &= \mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{C_1=x_1}\times\ldots\times\mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{C_T=x_T}|\mathcal{F}_{T-1}\right]\right]\\ &= (1+r)\mathbb{E}^{\mathbb{Q}}\left[\mathbf{1}_{C_1=x_1}\times\ldots\times\mathbf{1}_{C_{T-1}=x_{T-1}}\right]\\ &\cdots\\ &= (1+r)^T\\ &= \prod_{j\in[\![0,T]\!]}\mathbb{Q}\left[C_j=x_j\right].\end{aligned} \tag{6.1.3}$$

Let $\omega := (\omega_1,\cdots,\omega_T)$ in Ω and $x_t := C_t(\omega)$ for any t in $[\![0,T]\!]$. By independence of $C_1,\cdots,C_T$,

$$\mathbb{Q}[(\omega_1,\cdots,\omega_T)] = \mathbb{Q}\left[\cap_{j\in[\![0,T]\!]}C_j = x_j\right]$$

$$= \prod_{j \in [\![0,T]\!]} \mathbb{Q}\left[C_j = x_j\right],$$

hence we only need to specify for any j, $\mathbb{Q}\left[C_j = u\right]$. Taking expectation in (6.1.2) we deduce that:

$$u \times \mathbb{Q}\left[C_j = u\right] + d \times \left(1 - \mathbb{Q}\left[C_j = u\right]\right) = 1 + r,$$

from which we get that:

$$\mathbb{Q}\left[C_j = u\right] = \frac{1 + r - d}{u - d}.$$

Hence, the variables C_j are iid under $\mathbb{Q}$ and the previous relation together with (6.1.3) prove that $\mathbb{Q} = \mathbb{P}^*$ which concludes the proof. □

6.2 Dynamic programming principle

In this section we give an algorithm which enables one to compute the price of some any European contingent claim on the market. Consider a contract whose payoff G at time T is a contingent claim of European type of the form:

$$h = g(S_T),\ \mathbb{P} - a.s..$$

We aim in computing the price G_0 at time $t = 0$ of this claim with:

$$\rho_0 = (1 + r)^{-T} \mathbb{E}^{\mathbb{P}^*}[g(S_T)],$$

where $\mathbb{P}^*$ is the unique EMM on the market given in Proposition 6.1.3. To do this we will construct an induction in time on each time step. More precisely,

Theorem 6.2.1 (Dynamic programming)**.** *For t in $[\![0, T]\!]$, recall that the price ρ_t of h at time t is given by:*

$$\rho_t = (1 + r)^{-(T-t)} \mathbb{E}^{\mathbb{P}^*}[h|\mathcal{F}_t].$$

The family of prices $(G_t)_{t \in [\![0,T]\!]}$ satisfy:

$$\rho_t = g_t(S_t)(1 + r)^{-(T-t)},$$

where the functions g_t are explicitly determined using the following backward induction:

$$\begin{cases} g_T(x) := g(x), \\ \\ g_t(x) := g_{t+1}(xu)p^* + g_{t+1}(xd)(1 - p^*), \quad t \in \{0, \cdots, T_1\}, \quad x \in \mathcal{V} \end{cases}$$

where $\mathcal{V}$ denotes the possible values of the price S.

In addition, a replicating strategy π is given by $\pi_t = (1 + r)^{-(T-t)} \varphi_t(S_{t-1})$ for any t with

$$\varphi_t(x) := \frac{g_t(xu) - g_t(xd)}{x(u - d)}, \quad x \in \mathcal{V}, \quad t \in \{1 \ldots, T\}.$$

Proof. The proof relies on the Markov structure of the pair $(S, (\mathcal{F}_t)_t)$. We proceed by backward induction. Assume that Relation (6.2.1) is true at a given time $t \geq T$. Now we compute ρ_{t-1}.

$$\rho_{t-1} = (1+r)^{-(T-t+1)}\mathbb{E}^{\mathbb{P}^*}[g(S_T)\mathcal{F}_{t-1}] = (1+r)^{-1}\mathbb{E}^{\mathbb{P}^*}[\rho_t|\mathcal{F}_{t-1}].$$

Using the induction hypothesis, we have that:

$$\mathbb{E}^{\mathbb{P}^*}[\rho_t|\mathcal{F}_{t-1}] = (1+r)^{-(T-t)}\mathbb{E}^{\mathbb{P}^*}[g_t(S_t)|\mathcal{F}_{t-1}] = (1+r)^{-(T-t)}\mathbb{E}^{\mathbb{P}^*}[g_t(S_{t-1}C_t)|\mathcal{F}_{t-1}].$$

Using the Markov property,

$$\rho_{t-1} = (1+r)^{-(T-(t-1))}g_{t-1}(S_{t-1}),$$

where

$$g_{t-1}(x) := \mathbb{E}^{\mathbb{P}^*}[g_t(xC_t)] = g_t(xu)p^* + g_t(xd)(1-p^*).$$

We turn to the expression of the hedging strategy π. By definition, we have that $\rho_t = X_t^{(x,\pi)}$ for any t, where $x = \rho_0$. Hence:

$$\begin{aligned}(1+r)^{-(T-t)}g_t(S_{t-1}u)\mathbf{1}_{C_t=u} &= (\pi_t S_{t-1}(u-1) + X_{t-1}^{(x,\pi)})\mathbf{1}_{C_t=u}\\ (1+r)^{-(T-t)}g_t(S_{t-1}d)\mathbf{1}_{C_t=d} &= (\pi_t S_{t-1}(d-1) + X_{t-1}^{(x,\pi)})\mathbf{1}_{C_t=d}.\end{aligned}$$

Taking condition expectation $\mathbb{E}^{\mathbb{P}^*}[\cdot|\mathcal{F}_{t-1}]$ in both equalities and making the difference of the first one and of the second one, on gets

$$\pi_t = (1+r)^{-(T-t)}\frac{g_t(S_{t-1}u) - g_t(S_{t-1}d)}{S_{t-1}(u-d)}.$$

The conclusion follows by the Markov property. □

Remark 6.2.2. *For the Dynamic Programming equation to be valid one only needs the price process S to be a Markov process with respect to the filtration (which is much more general than the CRR model). For the CRR model, the Markov property is just a consequence of the independence of the random variables $C_1, \cdots, C_T$ under $\mathbb{P}^*$.*

More material about the CRR model can be dound in [11].

Chapter 7

Pricing and hedging in incomplete markets: the superhedging approach

In all this chapter, we work under the probabilistic setup of Chapter 5, *i.e.* we consider a finite filtered probability space $(\Omega, \mathcal{F}_T, (\mathcal{F}_t)_{t\in\{0,\dots,T\}}, \mathbb{P})$.

One may wonder if most of financial markets are complete. In fact, for discrete markets it has been shown in [7] that a discrete time market is complete if and only if at any time t the conditional law of $S_t - S_{t-1}$ given $\mathcal{F}_{t-1}$ is discretely supported on a set containing at most the number of risky assets S plus 1. In other words, many markets are not complete and most of real financial markets are incomplete.

We now consider an incomplete market, which by Theorem 5.5.4 is equivalent to the fact that the set $\mathcal{P}$ of EMM is not a singleton. And by definition, on an incomplete market there exist financial claims h which cannot be replicated, that is,

$$\mathbb{P}[X_T^{(x,pi)} \neq h] > 0, \quad \forall (x,\pi) \text{ portfolio.}$$

However, as Ω is finite, it is always possible to find a portfolio (x,π) such that:

$$X_T^{(x,\pi)} \geq h, \quad \mathbb{P}-a.s..$$

We say that (x,π) *superreplicates* h.

Definition 7.1.1 (Superreplication price)**.** *Let h be a financial claim, that is an element of $L^0(\Omega, \mathcal{F}_T)$.*

(i) A portfolio (x,π) is called a superreplication strategy if:

$$X_T^{(x,\pi)} \geq h, \quad \mathbb{P}-a.s..$$

(ii) The superreplication price $s(h)$ of h is defined as:

$$s(h) := \inf\left\{x \geq 0,\ \exists \pi \text{ predictable },\ X_T^{(x,\pi)} \geq h,\ \mathbb{P}-a.s..\right\}.$$

In other words, the superreplication price is the smallest initial capital for which it is possible to find a strategy which superreplicate the claim. The main result of this section is the theorem below which characterizes the superreplication price.

Theorem 7.1.2. *Let h be a given financial claim. We have that:*

$$s(h) = \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\frac{h}{(1+r)^T}\right].$$

Proof. First note that since Ω is finite, the set $\Gamma := \left\{x \geq 0,\ \exists \pi \text{ predictable },\ X_T^{(x,\pi)} \geq h,\ \mathbb{P}-a.s..\right\}$ is not empty as it contains $\max_{\omega\in\Omega} h(\omega)$ (and then choose $\pi := 0$). Let x in Γ. Hence there exists a strategy π such that (x,π) superreplicates h. Let $\mathbb{Q}$ be an EMM. By definition, $\tilde{X}^{(x,\pi)}$ is a $\mathbb{Q}$-martingale which means that:

$$x = \mathbb{E}^{\mathbb{Q}}[\tilde{X}_T^{(x,\pi)}] \geq \mathbb{E}^{\mathbb{Q}}\left[\frac{h}{(1+r)^T}\right].$$

Hence,

$$x \geq \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\frac{h}{(1+r)^T}\right],$$

and

$$\inf\Gamma \geq \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\frac{h}{(1+r)^T}\right].$$

The other inequality is a way more technical and will need several intermediary results that are presented in a series of lemma listed immediately after this proof. These lemma bring insights that are important results and/or technics in mathematical finance which is our motivation to write them done in this Lecture Notes. As a consequence, the reader may skip them at first reading.

Consider the stochastic process:

$$U_t := \text{esssup}_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\tilde{h}|\mathcal{F}_t\right], \quad t\in[0,T],$$

where we set $\tilde{h} := \frac{h}{(1+r)^T}$ for simplicity. The notion of *essential supremum* (esssup) is defined in Appendix A. The scheme of the proof is the following: First we prove that U is a $\hat{\mathbb{Q}}$-supermartingale for any $\hat{\mathbb{Q}}$ element of $\mathcal{P}$. Then by uniform Doob-Meyer decomposition (see Lemma 7.1.3), there exists a predictable process π and a non-decreasing (adapted) process A with $A_0 = 0$ such that:

$$U_t = \tilde{X}_t^{(U_0,\pi)} - A_t, \quad \forall t\in[0,T],\ \mathbb{P}-a.s..$$

Hence, $\tilde{X}_T^{(U_0,\pi)} \geq U_T = \tilde{h}$, $\mathbb{P}$-a.s. and by definition $U_0 = \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\tilde{h}\right]$. Hence

$$U_0 = \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}\left[\tilde{h}\right] \geq \inf\Gamma,$$

which is the desired inequality. Thus it remains to establish that U is a $\hat{\mathbb{Q}}$-supermartingale for any $\hat{\mathbb{Q}}$ in $\mathcal{P}$. To this end fix $\hat{\mathbb{Q}}$ in $\mathcal{P}$ and t in $\{0,\dots,T-1\}$. We need to prove that[1]:

$$\mathbb{E}^{\hat{\mathbb{Q}}}[U_{t+1}|\mathcal{F}_t] \leq U_t.$$

To see this, assume that the essential supremum defining U_{t+1} is attained at a random variable, that is, $U_{t+1} = \mathbb{E}^{\mathbb{Q}^*}\left[\tilde{h}|\mathcal{F}_{t+1}\right]$ where $\mathbb{Q}^*$ is absolutely continuous with respect to $\hat{\mathbb{Q}}$ with density Z (where we assume that $Z_s = 1$ for $s \leq t+1$). Then, by tower property,

$$\mathbb{E}^{\hat{\mathbb{Q}}}[U_{t+1}|\mathcal{F}_t] = \mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{h}|\mathcal{F}_t] \leq U_t.$$

However there are no reasons for the essential supremum in the very definition of U_{t+1} to be attained. We now from Appendix A that if the family is directed upwards (see the definition in Appendix A) then the essential supremum will be approximated by a monotonic sequence of conditional expectations with respect to the initial measure which will allow us to make the same line of reasoning as above. We now describe the argument. By Lemma 7.1.5 (with $\mathbb{P}^* := \hat{\mathbb{Q}}$),

$$U_{t+1} = \operatorname{esssup}_{Z\in\mathcal{P}^{\hat{\mathbb{Q}}}_{t+1}} \mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{h}|\mathcal{F}_{t+1}].$$

Now we prove that the family of random variables $\left(\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{h}|\mathcal{F}_{t+1}]\right)_{Z\in\mathcal{P}_{t+1}}$ is directed upwards. To this end, let Z^1, Z^2 in $\mathcal{P}_{t+1}$ (see Definition (7.1.1) below) and let $B := \left\{\mathbb{E}^{\hat{\mathbb{Q}}}[Z^1_T\tilde{h}|\mathcal{F}_{t+1}] \geq \mathbb{E}^{\hat{\mathbb{Q}}}[Z^2_T\tilde{h}|\mathcal{F}_{t+1}]\right\}$ (which belongs to $\mathcal{F}_{t+1}$). We set:

$$Z_s := \begin{cases} 1, & \text{if } s \leq t+1, \\ Z^1_s\mathbf{1}_B + Z^2_s\mathbf{1}_{B^c}, & \text{if } s > t+1. \end{cases}$$

Note that Z is adapted and that by definition,

$$\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{h}|\mathcal{F}_{t+1}] \geq \mathbb{E}^{\hat{\mathbb{Q}}}[Z^i_T\tilde{h}|\mathcal{F}_{t+1}],\ i = 1,2.$$

[1] in full generality, we would need to prove that

$$\mathbb{E}^{\hat{\mathbb{Q}}}[U_{t+1}|\mathcal{F}_t] \leq U_s,$$

for any $s \leq t+1$. However, as the time is discrete this is a consequence of the previous fact. Indeed:

$$\begin{aligned} \mathbb{E}^{\hat{\mathbb{Q}}}[U_{t+1}|\mathcal{F}_t] - U_s &= \sum_{i=s}^{t} \mathbb{E}^{\hat{\mathbb{Q}}}[U_{i+1} - U_i|\mathcal{F}_s] \\ &= \sum_{i=s}^{t} \mathbb{E}^{\hat{\mathbb{Q}}}[\mathbb{E}^{\tilde{Q}}[U_{i+1} - U_i|\mathcal{F}_i]|\mathcal{F}_s] \\ &\leq 0. \end{aligned}$$

We prove that Z belongs to $\mathcal{P}_{t+1}$. It remains to prove that $\tilde{S}$ is a $\mathbb{Q}^Z$-martingale with $\frac{d\mathbb{Q}^Z}{d\hat{\mathbb{Q}}} := Z_T$. Let s in $\{0, \dots, T-1\}$. First assume that $s \leq t$. By definition of Z it holds that:

$$\begin{aligned}
&\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T|\mathcal{F}_s] \\
&= \mathbb{E}^{\hat{\mathbb{Q}}}[\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T|\mathcal{F}_{t+1}]|\mathcal{F}_s] \\
&= \mathbb{E}^{\hat{\mathbb{Q}}}[\mathbf{1}_B \mathbb{E}^{\mathbb{P}^*}[Z_T^1|\mathcal{F}_{t+1}] + \mathbf{1}_{B^c}\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T^2|\mathcal{F}_{t+1}]|\mathcal{F}_s] \\
&= 1.
\end{aligned}$$

We hence deduce using Bayes formula (see Lemma 7.1.4) that:

$$\begin{aligned}
&\mathbb{E}^{\mathbb{Q}^Z}[\tilde{S}_{s+1}|\mathcal{F}_s] \\
&= \mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{S}_{s+1}|\mathcal{F}_s] \\
&= \mathbb{E}^{\hat{\mathbb{Q}}}[\mathbb{E}^{\mathbb{P}^*}[Z_T\tilde{S}_{s+1}|\mathcal{F}_{t+1}]|\mathcal{F}_s] \\
&= \mathbb{E}^{\hat{\mathbb{Q}}}[\tilde{S}_{s+1}\mathbb{E}^{\mathbb{P}^*}[Z_T|\mathcal{F}_{t+1}]|\mathcal{F}_s] \\
&= \mathbb{E}^{\mathbb{P}^*}[\tilde{S}_{s+1}|\mathcal{F}_s] \\
&= \tilde{S}_s.
\end{aligned}$$

Now assume that $s \geq t+1$. In that case a direct computation proves that:

$$\mathbb{E}^{\mathbb{P}^*}[Z_T|\mathcal{F}_s] = Z_s.$$

We can use the definition of Z together with the fact that Z^1 and Z^2 belong to $\mathcal{P}_{t+1}$ to get:

$$\begin{aligned}
&\mathbb{E}^{\mathbb{Q}^Z}[\tilde{S}_{s+1}|\mathcal{F}_s] \\
&= (Z_s)^{-1}\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{S}_{s+1}|\mathcal{F}_s] \\
&= (Z_s)^{-1}\left(\mathbf{1}_B E^{\hat{\mathbb{Q}}}[Z_T^1\tilde{S}_{s+1}|\mathcal{F}_s] + \mathbf{1}_{B^c}E^{\hat{\mathbb{Q}}}[Z_T^2\tilde{S}_{s+1}|\mathcal{F}_s]\right) \\
&= (Z_s)^{-1}\left(\mathbf{1}_B Z_s^1\tilde{S}_s + \mathbf{1}_{B^c}Z_s^2\tilde{S}_s\right) \\
&= (Z_s)^{-1}Z_s\tilde{S}_s \\
&= \tilde{S}_s.
\end{aligned}$$

This finishes to prove that Z belongs to $\mathcal{P}_{t+1}$ and thus that the family $\left(\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T\tilde{h}|\mathcal{F}_{t+1}]\right)_{Z\in\mathcal{P}_{t+1}}$ is directed upwards. Hence, by Theorem A.1.1, there exists a sequence of elements $(Z^n)_n$ in $\mathcal{P}_{t+1}$ such that

$$U_{t+1} = \lim_{n\to+\infty}\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T^n\tilde{h}|\mathcal{F}_{t+1}],$$

and the sequence $\left(\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T^n\tilde{h}|\mathcal{F}_{t+1}]\right)_n$ is non-decreasing. As a consequence:

$$\mathbb{E}^{\hat{\mathbb{Q}}}[U_{t+1}|\mathcal{F}_t]$$

$$
\begin{aligned}
&= \lim_{n\to+\infty} \mathbb{E}^{\hat{\mathbb{Q}}}[\mathbb{E}^{\hat{\mathbb{Q}}}[Z_T^n \tilde{h}|\mathcal{F}_{t+1}]|\mathcal{F}_t]] \\
&= \lim_{n\to+\infty} \mathbb{E}^{\hat{\mathbb{Q}}}[Z_T^n \tilde{h}|\mathcal{F}_t] \\
&\leq \text{esssup}_{Z\in\mathcal{P}_t} \mathbb{E}^{\hat{\mathbb{Q}}}[\tilde{h}|\mathcal{F}_t] \\
&= U_t,
\end{aligned}
$$

where we have used Lemma 7.1.5 and the fact that by definition $\mathcal{P}_{t+1} \subset \mathcal{P}_t$. □

The next result is known under the name of *uniform Doob-Meyer decomposition* and its proof can be found in [4, Theorem 7.5] (we omit the proof here as we consider here to be too technical).

Lemma 7.1.3 (Uniform Doob-Meyer decomposition)**.** *Let a process U which is a $\mathbb{Q}$-martingale for any $\mathbb{Q}$ in $\mathcal{P}$. Then there exists a predictable π and a non-decreasing adapted process A with $A_0 = 0$ such that:*

$$
U_t = \tilde{X}_t^{(U_0,\pi)} - A_t, \quad \forall t \in [0,T],\ \mathbb{P}-a.s..
$$

Lemma 7.1.4 (Bayes formula)**.** *Let Z be a poisitive $\mathbb{P}$-martingale with $Z_0 = 1$. We set $\mathbb{Q}$ the measure equivalent to $\mathbb{P}$ with $\frac{d\mathbb{Q}}{d\mathbb{P}} := Z_T$. Then for any $\mathcal{F}_T$-measurable random variable h, and for any t in $\{1,\dots,T\}$, it holds that:*

$$
\mathbb{E}^{\mathbb{Q}}[h|\mathcal{F}_t] = \frac{\mathbb{E}[h|\mathcal{F}_t]}{Z_t}.
$$

Proof. Left as an exercise. □

Lemma 7.1.5. *Let $\mathbb{P}^*$ in $\mathcal{P}$. For any t in $\{0,\dots,T\}$ we set*

$$
\mathcal{P}_t^{\mathbb{P}^*} := \{Z = (Z_s)_{s\in\{0,\dots,T\}},\ Z \text{ is a } \mathbb{P}^* - martingale,\ Z_s = 1, s \leq t\}. \tag{7.1.1}
$$

Let V be a $\mathcal{F}_T$-measurable random variable. We have that:

$$
esssup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}[V|\mathcal{F}_t] = esssup_{Z\in\mathcal{P}_t} \mathbb{E}^{\mathbb{P}^*}[Z_T V|\mathcal{F}_t].
$$

Proof. Obviously every element Z in $\mathcal{P}_t$ defines a measure $\mathbb{Q}$ in $\mathcal{P}$ by $\frac{d\mathbb{Q}}{d\mathbb{P}^*} := Z_T$ so that:

$$
\text{esssup}_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}[V|\mathcal{F}_t] \geq \text{esssup}_{Z\in\mathcal{P}_t} \mathbb{E}^{\mathbb{P}^*}[Z_T V|\mathcal{F}_t].
$$

Conversely take $\mathbb{Q}$ in $\mathcal{P}$. There exists $\tilde{Z}_T$ such that $\tilde{Z}_T = \frac{d\mathbb{Q}}{d\mathbb{P}^*}$. We set: $Z_T := \frac{\tilde{Z}_T}{\tilde{Z}_t}$ and $Z_s := \mathbb{E}^{\mathbb{P}^*}[Z_T|\mathcal{F}_s]$ for $s \leq T$ so that Z is $\mathbb{P}^*$-martingale wich is constant up to time t as for any $s \leq t$,

$$
Z_s = \mathbb{E}^{\mathbb{P}^*}[\mathbb{E}^{\mathbb{P}^*}[Z_T|\mathcal{F}_t]|\mathcal{F}_s] = 1.
$$

This proves that the constructed process Z belongs to $\mathcal{P}_t$. Finally by Bayes formula,

$$
\mathbb{E}^{\mathbb{Q}}[V|\mathcal{F}_t] = \mathbb{E}^{\mathbb{P}^*}[V\tilde{Z}_T(Z_t)^{-1}|\mathcal{F}_t] = \mathbb{E}^{\mathbb{P}^*}[V Z_T|\mathcal{F}_t].
$$

Since for any $\mathbb{Q}$ in $\mathcal{P}$ we have associated an element Z in $\mathcal{P}_t$ which satisfy the equality above, we have that

$$\text{esssup}_{\mathbb{Q}\in\mathcal{P}}\mathbb{E}^{\mathbb{Q}}[V|\mathcal{F}_t] \leq \text{esssup}_{Z\in\mathcal{P}_t}\mathbb{E}^{\mathbb{P}^*}[Z_T V|\mathcal{F}_t],$$

which concludes the proof. □

Chapter 8

American options: pricing and hedging in incomplete markets

In all this chapter, we work under the probabilistic setup of Chapter 5, *i.e.* we consider a finite filtered probability space $(\Omega, \mathcal{F}_T, (\mathcal{F}_t)_{t\in\{0,\dots,T\}}, \mathbb{P})$.

8.1 Optimal stopping and Snell envelop

For any $0 \leq s \leq t \leq T$ we let:

$$\mathcal{T}_{s,t} := \{\tau \text{ stopping time }, \ s \leq \tau \leq t, \ \mathbb{P} - a.s..\}$$

the set of stopping times between s and t.

The optimal stopping problem consists of the following optimization problem. Given H an adapted stochastic process, an element $\mathbb{P}^*$ in $\mathcal{P}$, and t we consider:

$$V(t) := \text{esssup}_{\tau\in\mathcal{T}_{t,T}} \mathbb{E}^*[H_\tau|\mathcal{F}_t]. \tag{8.1.1}$$

One question is then to see if one can find an optimal stopping time. The key tool for studying this problem turns to be the Snell envelop.

Definition 8.1.1. *We define the Snell envelop (with respect to $\mathbb{P}^*$) of a given H be an adapted process, as the adapted process ν with:*

$$\begin{cases} \nu_T := H_T, \\ \nu_t := \max\left(H_t, \mathbb{E}^*[\nu_{t+1}|\mathcal{F}_t]\right), \quad t \leq T-1. \end{cases}$$

Lemma 8.1.2. *The Snell envelop ν of H is the smallest $\mathbb{P}^*$-supermartingale above H.*

Proof. By its very definition, ν dominates H and is a $\mathbb{P}^*$-supermartingale. We prove the minimality property that is that ν is smaller than any supermartingale that dominates H. Let $\hat{\nu}$ such a supermartingale. We prove the result by induction again. The property is

clearly true at T. Assume it holds at a time $t \leq T$. Then using the induction assumption, the supermartingale property of $\hat{\nu}$ and the fact that $\hat{\nu}$ dominates H, it holds that:

$$\begin{aligned}\nu_{t-1} &= \max\left(H_{t-1}, \mathbb{E}^*[\nu_t|\mathcal{F}_{t-1}]\right)\\ &\leq \max\left(H_{t-1}, \mathbb{E}^*[\hat{\nu}_t|\mathcal{F}_{t-1}]\right)\\ &\leq \max\left(H_{t-1}, \hat{\nu}_{t-1}\right)\\ &\leq \hat{\nu}_{t-1}.\end{aligned}$$

So the proof is complete. □

Lemma 8.1.3. *For t in $\{0, \leq T\}$, we set:*

$$\tau_t^* := \inf\{s \geq t,\ H_s = \nu_s\} \wedge T. \tag{8.1.2}$$

Then $(\nu_{s\wedge\tau_t^*})_{s\geq t}$ is a martingale.

Proof. Let t in $\{0, \ldots, T-1\}$ and $s \geq t$. We have:

$$\begin{aligned}\nu_{(s+1)\wedge\tau_t^*} - \nu_{s\wedge\tau_t^*} &= \mathbf{1}_{\{\tau_t^* \geq s+1\}}(\nu_{s+1} - \nu_s)\\ &= \mathbf{1}_{\{\tau_t^* \geq s+1\}}(\nu_{s+1} - \mathbb{E}^*[\nu_{s+1}|\mathcal{F}_s]),\end{aligned}$$

which entails that:

$$\mathbb{E}^*[\nu_{(s+1)\wedge\tau_t^*} - \nu_{s\wedge\tau_t^*}|\mathcal{F}_s] = 0,$$

as $\{\tau_t^* \geq s+1\} = \{\tau_t^* \leq s\}^c \in \mathcal{F}_s$. □

Now we come back to the initial stopping problem (8.1.1).

Theorem 8.1.4. *With the notations above, we have that:*

$$\nu_t = \mathbb{E}^*[H_{\tau_t^*}|\mathcal{F}_t], \quad \forall t \in [\![0, T]\!],\ \mathbb{P}-a.s.,$$

where τ_t^* is defined in (8.1.2)*. In addition,*

$$V(t) = \nu_t, \quad \forall t \in [\![0, T]\!],\ \mathbb{P}-a.s..$$

Proof. Let t in $[\![0, T]\!]$. We have by Lemma 8.1.3 that:

$$\nu_t = \mathbb{E}^*[\nu_{T\wedge\tau_t^*}|\mathcal{F}_t] = \mathbb{E}^*[H_{\tau_t^*}|\mathcal{F}_t].$$

Hence, the conclusion follows if we prove that $\nu = V$. As ν is a supermartingale, it holds that:

$$\nu_t \leq \mathbb{E}^*[\nu_{T\wedge\tau_t^*}|\mathcal{F}_t] \leq \mathbb{E}^*[H_{\tau_t^*}|\mathcal{F}_t] \leq V(t).$$

Let τ in $\mathcal{T}_{t,T}$. As ν is a supermartingale, it holds that:

$$\nu_t \geq \mathbb{E}^*[\nu_\tau|\mathcal{F}_t] \geq \mathbb{E}^*[H_\tau|\mathcal{F}_t].$$

So $\nu_t \geq V(t)$. Finally, we have that:

$$\nu_t \leq \mathbb{E}^*[H_{\tau_t^*}|\mathcal{F}_t] \leq V(t) = \nu_t.$$

□

Remark 8.1.5. *One may be a little bit puzzled by the fact that in the proof above we have established that* $\nu_t = \mathbb{E}^*[\nu_{T\wedge\tau_t^*}|\mathcal{F}_t] = \mathbb{E}^*[H_{\tau_t^*}|\mathcal{F}_t]$ *but after that, we have only used inequalities. We have done this to point out that the proof can be carried out if Lemma 8.1.3 is replaced by only a supermartingale property. This remark is useful to show the counterpart of Theorem 8.1.4 in the pricing and hedging problem of American options in incomplete markets.*

8.2 American options in complete markets

In case of a European option the buyer can exercise her option only once at a given time. An American option is a contract where the buyer is allowed to exercise her option at any time before maturity (or even not to exercise it).

Definition 8.2.1 (American contingent claims)**.** *An American contingent claim is an adapted process* $(G_t)_{t\in[\![0,T]\!]}$*, where each random variable* G_t *represents the payoff of the contract if the holder exercise at time* t*.*

To make things as simple as possible, consider as an example an American call option on a stock S with strike K. If the agent chooses to exercise at a time t in $[\![0,T]\!]$, then she will receive the payoff $G_t := (S_t - K)_+$. When $t = T$, G_T is the payoff of a classical European call option. If the seller wants to be able to hedge this contract, she must build a portfolio (x,π) such that:

$$X_t^{(x,\pi)} \geq G_t,\ \forall t \in [\![0,T]\!],\ \mathbb{P}-a.s.. \tag{8.2.1}$$

In addition, as the option may not be exercised, the additional requirement:

$$X_T^{(x,\pi)} \geq 0,\ \mathbb{P}-a.s.. \tag{8.2.2}$$

must be in force. In addition, this requirement is not strong enough, as the buyer may not choose a deterministic time of exercise but a random one. That is, Relation (8.2.1) have to be replaced by:

$$X_\tau^{(x,\pi)} \geq G_\tau,\ \forall \tau \text{ stopping time },\ \mathbb{P}-a.s.. \tag{8.2.3}$$

The first question is then :

"Can we find a portfolio (x,π)*, such that the two requirements* (8.2.2)-(8.2.3) *are satisfied?"*

From now on, $(G_t)_{t\in[\![0,T]\!]}$ denotes an adapted process where G_t is the payoff received by the holder of the option if she exercise at time t. Considering discounted values and taking expectation in (8.2.3) for a stopping time τ in $\mathcal{T}_{0,T}$ gives:

$$x \geq \mathbb{E}^*[\tilde{G}_\tau],$$

and so

$$x \geq \sup_{\tau\in\mathcal{T}_{0,T}} \mathbb{E}^*[\tilde{G}_\tau].$$

The question then becomes :

"Can we find a strategy π such that the portfolio ($x := \sup_{\tau\in\mathcal{T}_{0,T}} \mathbb{E}^*[\tilde{G}_\tau], \pi$) can hedge the option?"

To answer this question, consider, the Snell envelop ν of the process of discounted payoffs $\tilde{G} := (\tilde{G}_t)_{t\in[\![0,T]\!]}$. If such a portfolio exists, then we have that

$$x = \tilde{X}_0^{(x,\pi)} = \nu_0 = \sup_{\tau\in\mathcal{T}_{0,T}} \mathbb{E}^*[\tilde{G}_\tau].$$

The main difficulty is that G_T is not exactly the payoff of the contract since it depends on the exercise time of the buyer, so we then have a process of payoffs. This is one of the main differences between contract which are exercised at a fixed time and American contracts. However, one can still tries to figure out what is the hedge in this type of contract. We know that $\tilde{X}^{(x,\pi)}$ is a martingale. And we know as well that to be able to face the payment at any stopping time τ it should dominate $\tilde{G}$ at any of these times. Recall that we call ν the Snell envelop of $\tilde{G}$. As it is a supermartingale, by Doob-Meyer decomposition, there exists a unique martingale M and a unique non-decreasing predictable process A with $A_0 = 0$ such that:

$$\nu = M - A.$$

The completeness assumption of the market implies that there exists a unique pair $(\bar{x}, \bar{\pi})$ such that:

$$(1+r)^T M_T = X_T^{(\bar{x},\bar{\pi})}.$$

Hence, $\tilde{X}^{(\bar{x},\bar{\pi})}_{\cdot\wedge\tau_0^*}$ is the martingale $M_{\cdot\wedge\tau_0^*}$ and:

$$\bar{x} = \mathbb{E}^*[\tilde{X}^{(\bar{x},\bar{\pi})}_{T\wedge\tau_0^*}] = \mathbb{E}^*[M_{T\wedge\tau_0^*}] = \nu_0 = x.$$

As a consequence the choice $\pi := \bar{\pi}$ provides a hedge.

We can go a little bit further in the description above. We feel that for American options the notion of replication (and of portfolio itself) needs to be extended. Coming back to the decomposition above, we have proved that:

$$\tilde{X}_t^{(x,\pi)} = \nu_t + A_t.$$

So if $t \le \tau_0^*$, $A_t = 0$, but after the optimal exercise time τ_0^*, $A > 0$ and $\nu \ge \tilde{G}$, so we can face the payment of $\tilde{G}$ but in addition we can allow for some extra cash A which can be seen as a consumption term. This in fact calls for an extension of the notion of portfolio where one allows for an extra component which is the consumption. In that case, the notion of self-financing portfolio needs to be extended but we will not go into this direction in this course (see *e.g.* [9, 2]).

8.3 American options in incomplete markets

In incomplete market, we can adapt the previous arguments to derive a superreplication price for an American option payoff. Roughly, speaking we have seen that the American framework requires the introduction of the Snell envelop, and the incomplete feature of the market needs to consider supermartingales with respect to a family of measures. The American options pricing and hedging problem requires to combine these two types of techniques. We state below the main result in this context and we refer to [2, Section 8.3.3] for a detailed proof. However, we recommend to the reader to try to adapt the arguments of the previous section to prove this theorem.

Theorem 8.3.1. *Let $\mathcal{P}$ the set of EMM and $(G_t)_{t\in[\![0,T]\!]}$ the payoff associated to an American option. Then the superhedging price ν_0 of this claim is given by:*

$$\nu_0 = \sup_{\tau\in\mathcal{T}_{0,T}} \sup_{\mathbb{Q}\in\mathcal{P}} \mathbb{E}^{\mathbb{Q}}[\tilde{G}_\tau] = \sup_{\mathbb{Q}\in\mathcal{P}} \sup_{\tau\in\mathcal{T}_{0,T}} \mathbb{E}^{\mathbb{Q}}[\tilde{G}_\tau].$$

In addition, ν_0 coincides with the value at $t = 0$ of the stochastic process (called the superior Snell envelop of the discounted payoffs $\tilde{G}$)

$$\begin{array}{l} \nu_T := \tilde{G}_T \\ \nu_t := \max\left(\tilde{G}_t, esssup_{\mathbb{Q}\in\mathcal{P}}\mathbb{E}^{\mathbb{Q}}[\tilde{\nu}_{t+1}|\mathcal{F}_t]\right), \quad t \leq T-1 \end{array}.$$

ν is the smallest $\mathcal{P}$-supermartingale[1] *which dominates $\tilde{G}$. By uniform Doob-Meyer decomposition (see Lemma 7.1.3) there exists a predictable non-decreasing process A with $A_0 = 0$ and a predictable process π such that:*

$$\nu = \tilde{X}^{(\nu_0,\pi)} - A.$$

The process π provides an hedging strategy.

[1] which means that it is a $\mathbb{Q}$-supermartingale for any $\mathbb{Q}$ in $\mathcal{P}$

Part III

Exercise sheets

Chapter 9

Exercise sheet 1: corporate finance

Exercise 1 :[ENSO (El Niño Southern Oscillation)] ENSO is the name given to the pseudo-random natural phenomenon (which occurs approximately every 3 to 8 years) also known as El Niño which consists of a sudden increase of the temperature at the surface of East Pacific Ocean. This phenomenon entails for instance an increase of rainfall nearby South America. This consequence is profitable to some farmers (those who cultivate cotton for instance) in dry regions. In the same time, the increase of surface ocean leads to less fish catches for fishermen in this area. Thus El Niño creates a situation where two types of economic agents: farmers Fa and fishermen Fi are exposed to an opposite risk.

The goal of this exercise, is to propose a very simple contract which enables the farmers and the fishermen to reduce their exposure to El Niño.

1) Let e and e' be the two events:

$$e = \{\text{ENSO occurs the year } X\}, \quad e' = \{\text{ENSO does not occurs the year } X\}$$

The wealth (in a given unit of numéraire) of the agents Fa are Fi are given in the table below.

	Fa	Fi
e	3	2
e'	1	4

Plot in a graph (whose horizontal axis represents the wealth in case of e and whose vertical axis represents the wealth in case of e') the points Fa and Fi. Plot as well the bisectrix and give an interpretation of the points on it.
2) The 1st of december of year X, Fa and Fi subscribe to the following contract:
"On December 31st of the year X, Fa pays $\frac{1}{2}$ to Fi in case of e; otherwise Fi pays $\frac{1}{2}$ to A." We denote by Fa^c and Fi^c the new coordinates of the wealths of Fa and Fi with this contract. Add these points Fa^c and Fi^c on the graph and give an interpretation of the positions compared to the bisectrix.

3) We now replace the table of 1) by the following one:

	Fa	Fi
e	3	1
e'	1	3

Design a contract between Fa and Fi so that the risk of Fa and Fi vanishes.

Note that this simple exercise provides an example of a securitization contract where no cash flows are exchanged at the signature of the contract but only at maturity. Note also that we make here a very elementary analysis of this problem but the real market based on this situation calls for a much deeper analysis which goes beyond the objectives of this course.

Exercise 2 :[Einstein's rule] Let r be an annual compound interest rate. We invest a capital S at time 0 with respect to r and denote by n the minimal number of years after which the capital is at least equal to $2S$.
1) Compute n.
2) Assume that $r \leq 0.05$. Prove that n can be approximated by

$$n \approx \frac{69,31}{\tau}, \text{ with } \tau := r \times 100.$$

This approximation is known under the name of *Einstein's rule* with $69,31$ replaced by 72.

Exercise 3 :[Price of a bond] Consider an annual interest rate $r = 3\%$. Compute the (arbitrage) price of a bond with par 10 000€ reimbursed at par with maturity 5 years and with coupon rate z with:

(i) $z = 1\%$

(ii) $z = 3\%$

(iii) $z = 5\%$

In light of your results give an explanation to the classical economic concept according to which interest rates move opposite to bond prices.

Exercise 4 :[Arbitrage price] Consider a financial market composed of 4 financial products: A, B, C, D whose cash flows are given below (the unit of time is the year):

Date	0	1	2	3
A	−100	150	10	30
B	−0.80	1	0	0
C	−0.70	0	1	0
D	−0.50	0	0	1

1) What are B, C, D?
2) Is there any anomaly on this market?
3) Can we modify the price of product A so that the anomaly detected in 2) disappears?
4) We have seen in Chapter 3 that the arbitrage price of A should be:

$$\sum_{i=1}^{n} \frac{F_i}{(1+z(t_i))^{t_i}}.$$

Explain all the quantities in this formula. With this formula at hand, compute the price of A and compare it to the price obtained in 3).
5) Construct the yield curve given by B, C, D (using a linear interpolation).
6) Using the yield curve, give an estimation of $z(2.5)$. Explain the meaning of this value.
7) Give the price (at $t = 0$) of a z.-c. bond on this market with maturity 2 years and a half.

Exercise 5 :[Forward Rate] Consider $0 \leq t \leq T \leq S$ and a market where z.-c.bonds of maturities T, S and $S-T$ are available (we do not assume that these bonds are related to an interest rate). At time t, we want to invest one unit of numéraire on the period $[T, S]$. To do so, we use the following strategy: At t, one sells a T-bond and reinvests in a S-bond to result in a zero net investment.
1) Summarize in a table the cash flows generated by this strategy at times t, T, S. In particular what is the balance at time S.
2) Show that the rate $F(t, T, S)$ defined as:

$$F(t,T,S) = \frac{1}{S-T}\left(\frac{p(t,T)}{p(t,S)} - 1\right),$$

is a simple rate associated to the strategy under consideration.
The rate $F(t, T, S)$ is called the *simple forward rate for* $[T, S]$ *prevailing at* t.

Exercise 6 : Let $0 \leq t \leq T \leq S$. We denote by $p(a, b)$ the price of a z.-c. bond subscribed at time a and with maturity b with $a, b \in \{t, T, S\}$ and $a \leq b$. We assume that these bonds are relative to a deterministic short rate.
1) Show that $p(t, S) = p(t, T) \times p(T, S)$.
2) Assume in addition that the spot rate is constant and equal to $e^r - 1$ with $r > 0$. Express $p(t, T)$ in terms of r.
3) Now we consider a deterministic mapping $r := (r_s)_{s\in[0,T]}$. Let t in $[0, T]$, $n \geq 1$ and $t_i := t + \frac{i(T-t)}{n}$ for $i \in \{0, \cdots, n\}$. We assume that during the period $[t_i, t_{i+1}]$ the annual interest rate of the riskless asset is $e^{r_i} - 1$. Compute the value of the price $p^n(t, T)$ associated to the z.c. with rates $(e^{r_i} - 1)_i$. Prove that the limiting price $p(t, T)$ with respect to r continuously compounded with respect to r is:

$$p(t,T) = e^{-\int_t^T r_s ds}.$$

Compare with the expression obtained in 2) for $r_s = r$.

Exercise 7 :[Floatting rates and swaps] Let $0 < T_1 < T_2 < \ldots < T_n$. We consider a bond with par N reimbursed at par with coupons $c_1, \ldots, c_n$ in $T_1, \ldots, T_n$. We assume that $T_{i+1} - T_i = 1$ year.
1) Assume that the coupons are relative to a deterministic annual rate z. Compute the price p in $t = 0$ of the bond in terms of the prices $p(0, T_i)$ of the z.-c. with maturity T_i ($i \in \{1, \cdots, n\}$).
2) We now assume that the coupons are computed with respect to floatting (that is non deterministic) rates. More precisely, we assume that:

$$c_i := L(T_{i-1}, T_i)N,$$

where $L(T_{i-1}, T_i)$ is the simple interest rate associated to a z.-c. bond with price $p^L(T_{i-1}, T_i)$ on $[T_{i-1}, T_i]$ ($i \in \{1, \ldots, n\}$). For simplicity, we assume that $N = 1$.

a) Express $L(T_{i-1}, T_i)$ in terms of $p^L(T_{i-1}, T_i)$. Deduce the value of each coupon c_i in terms of $p^L(T_{i-1}, T_i)$.

b) How much money is needed at time $t = 0$ to get c_i at time T_i?

c) Compute the price of the bond.
3) We now consider a *swap*: at each time T_i, the buyer of the interest rate swap pays zN and receives $L(T_{i-1}, T_i)N$ (recall that $N = 1$).

a) How much money the buyer must invest in this market at time $t = 0$ to ensure the payment of the net cash flow at a given time T_i.

b) Deduce from the previous question the value of the swap at time $t = 0$.

c) We call the *forward swap rate* the vale z above which makes the price computed in the previous question equals to 0 (we denote it by z^*). Prove that

$$z^* = \frac{1 - p^L(0, T_n)}{\sum_{i=1}^n p^L(0, T_i)}.$$

d) Rewrite z^* as:

$$z^* = \sum_{i=1}^n w_i F(0, T_{i-1}, T_i),$$

where w_i are weights to be determined and $F(0, T_{i-1}, T_i)$ denotes the forward rate defined in Exercise 5.

Note that if $n = 1$ such a swap is called a *Forward Rate Agreement*.

Chapter 10

Exercise sheet 2: discrete time financial markets

Exercise 8 :[Self-financing portfolio] Using notations of Section 5.3, prove that a predictable processes (π, π^0) define a self-financing portfolio if and only if

$$\pi_t \cdot S_t + \pi_t^0 S_t^0 = \pi_{t+1} \cdot S_t + \pi_{t+1}^0 S_t^0, \quad \forall t \geq 1.$$

Deduce from Relations (5.3.1) and (5.3.2) that

$$\Delta_t X_t = (\pi_t, \pi_t^0) \cdot \Delta_t(S_t, S_t^0).$$

Exercise 9 :[Pricing of a forward contract] We consider a complete financial market as in Chapter 5 with only 1 risky asset S. Let G_T denote the payoff of a forward contract on S with maturity T and strike K.
1) Recall the form of G_T.
2) Prove that the arbitrage price G_0 of the contract is given by:

$$G_0 = S_0 - \frac{K}{(1+r)^T}.$$

Exercise 10 :[Call-Put parity] We consider a complete financial market as in Chapter 5 with only 1 risky asset S. We consider a European call (respectively a put) option on S with maturity T and strike K with payoff C_T (respectively P_T) at maturity.
1) Recall the forms of C_T and P_T.
2) Give the form of the prices C_t and P_t at time t of these contracts for any time t.
3) Study the mapping:

$$(x-K)_+ - (K-x)_+, \quad x \in \mathbb{R}.$$

Deduce the *Call-Put parity* relation:

$$P_0 + S_0 = C_0 + \frac{K}{(1+r)^T}. \tag{10.0.1}$$

Exercise 11 :[Pricing in the Cox-Ross-Rubinstein model] Consider the market defined in Chapter 6.

1) Assume that $p = 0.9$, $r = 0.01$, $d = 0.5$ and $u = 2$, $S_0 = 100$, $T = 3$.

a) Is the market arbitrage free?

b) Is the market complete?

c) What are the EMM?

d) Compute the price at time 0 of a European call option with strike $K = 120$, and deduce the price of the put.

2) We consider general data p, r, d, u, S_0, T such that the market is arbitrage free. The aim of this question is to price a *lookback option*. A lookback option is a contract where the payoff G_T of the buyer at time T is given by:

$$G_T := S_T - \underline{S}_T, \text{ with } \underline{S}_t := \min_{0 \le s \le t} S_s.$$

a) Prove that if $d \geq 1$ then the price G_0 at time 0 of the option is equal to the price at $t = 0$ of a European call option with strike S_0.

b) From now, one we assume that $d < 1$. Show that a dynamic programming equation for the lookback option as derived in Section 6.2 for European claims holds true. More precisely prove that:

$$G_t := (1+r)^{-(T-t)} \mathbb{E}^{\mathbb{P}^*}[G_T | \mathcal{F}_t] = (1+r)^{-(T-t)} g_t(S_t, \underline{S}_t), \quad \forall t$$

where

$$g_T(x, y) = x - y$$
$$g_t(x, y) = g_{t+1}(xu, y)p^* + g_{t+1}(xd, \min(y, xd))(1 - p^*).$$

c) We now assume, throughout this exercise, that $d = \frac{1}{u}$ and that $d < 1$. Show that the dynamic programming equation rewrites as:

$$g_T(x, y) = x - y$$
$$g_t(x, y) = \begin{cases} g_{t+1}(xu, y)p^* + g_{t+1}(xd, y)(1 - p^*), & \text{if } x > y \\ g_{t+1}(xu, y)p^* + g_{t+1}(xd, yd)(1 - p^*), & \text{if } x = y. \end{cases}$$

d) Give a dynamic programming equation for the hedging strategy π of the contract.

e) Compute the price at time 0 of the lookback option with the data of 1).

Exercise 12 :[Trinomial market model] We consider a financial market as described in Chapter 5, with one risky asset S whose dynamic is given by:

$$S_t = S_{t-1} C_t, \quad t \in \{1, \cdots, T\},$$

with C_t iid random variables given by:

$$\mathbb{P}[C_t = u] = p; \ \mathbb{P}[C_t = d] = q; \ \mathbb{P}[C_t = 1] = 1 - (p + q),$$

with $0 < d < 1 < u$ and p, q are real numbers in $(0,1)$. We consider the filtration $\mathbb{F} := (\mathcal{F}_t)_{t\in[0,T]}$ with $\mathcal{F}_t := \sigma(C_1, \cdots, C_t)$. The riskless bond S^0 is given by:

$$S_t^0 := (1+r)^t,$$

with $r \geq 0$.
1) Describe the underlying probability space on this market.
2) Exhibit two AOs if the condition $d < 1 + r < u$ is not satisfied (one AO for each inequality).
3) From now, we assume now that $d < 1 + r < u$ is in force. Let $\mathbb{P}^*$ be an equivalent martingale measure. Prove that $C_1, \cdots, C_T$ are independent with respect to $\mathbb{P}^*$. Deduce that $\mathbb{P}^* = \mu^{\otimes T}$, with μ a probability measure on the underlying probability space of the one-period trinomial model.
4) Set $\mathbb{P}^*$ the probability measure defined by:

$$\mathbb{P}^*[C_t = u] = p^*;\ \mathbb{P}^*[C_t = d] = q^*;\ \mathbb{P}^*[C_t = 1] = v^*,$$

where p^*, q^*, v^* are three real numbers in $(0,1)$. Give a sufficient on v^* which ensures $\mathbb{P}^*$ to be an EMM.
5) Assume (just for this question) that $T = 1$, $p^* = q^* = v^* = \frac{1}{3}$, $r = 0.07$, $u = 2$ and $d = \frac{1}{2}$. Can we replicate the payoff of a European call option on S with strike $K = \frac{3}{2}$ and maturity $T = 1$ on this market?
6) Is the trinomial market complete in general?
7) In the setting of Question 5), give the superhedging price of the European call option on S with strike $K = \frac{3}{2}$ and maturity $T = 1$.

Part IV

Appendix, Index, English/French Glossary and Bibliography

Appendix A

Essential supremum

Consider a family of random variables $(X_i)_{i\in I}$ (defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$) indexed by a set I. We would like to stress here that I is not assumed to be countable. What we aim at is to define a new random variable $\bar{X}$ such that:

(i) $X_i \leq \bar{X}$, $\mathbb{P}$-a.s.. for any i in I

(ii) If there is a random variable Z such that $X_i \leq Z$, $\mathbb{P}$-a.s.. for any i in I, then $\bar{X} \leq Z$, $\mathbb{P}$-a.s..

A natural candidate seems to be $\bar{X}(\omega) := \sup_{i\in I} X_i(\omega)$ for any ω in Ω. However, as I may not be countable, the resulting random variable might not be measurable unless Ω is a finite set and $\mathcal{F} := \mathcal{P}(\Omega)$ (the set of subsets of Ω). However, the main result that we will not prove is to construct a unique random variable (so which is $\mathcal{F}$-measurable) which will satisfy (i) and (ii).

Theorem A.1.1. *There exists a unique random variable $\bar{X}$ (that is if there are two they are equal $\mathbb{P}$-a.s.) such that (i)-(ii) are in force. We call $\bar{X}$ the essential supremum of the family $(X_i)_{i\in I}$ and we denote it by*

$$esssup_{i\in I} X_i := \bar{X}.$$

If in addition the family is directed upwards, that is:

$$\forall (i,j) \in I^2,\ \exists k \in J,\ X_k \geq \max(X_i, X_j), \mathbb{P} - a.s.$$

then their exists a sequence $(i_n)_n$ in I such that $X_{i_n} \leq X_{i_{n+1}}$ for any n and:

$$\lim_{n\to+\infty} X_{i_n} = \bar{X}, \quad \mathbb{P} - a.s..$$

For the proof we refer to [4, Appendix A.5, Theorem A.32].

Remark A.1.2. *As noted above, if Ω is a finite set and if $\mathcal{F} = \mathcal{P}(\Omega)$ then*

$$esssup_{i\in I} X_i = \sup_{i\in I} X_i.$$

In this course we work with finite probability space, so there is no need to introduce the essential supremum. However, this simplification is only valid in the finite case and so we did not want to mislead the reader as in the general case the essential supremum is the right notion to solve the kind of problems we have encountered for instance in Section 7.

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Appendix B

Glossary

English	Français
Annual percentage rate	taux nominal annuel
Arbitrage	Arbitrage
Asset (Financial asset)	Actif (actif financier)
Bond	Obligation
Call Option (European call option)	Option d'achat (européenne)
Cashing	Encaissement
Compound interest rate	Taux d'intérêt composé
Contingent claim	Actif contingent
Cash flow	Flux de trésorerie
Coupon	Coupon
Commodity(ies)	Matière(s) première(s)
Compounded value	Valeur acquise
Compounding	Capitalisation
Derivative	Produit dérivé

English	Français
Disbursement	Décaissement
Discounted value	Valeur actualisée
Discounting	Actualisation
Dynamic Programming Principle	Principe de la programmation dynamique
Effective annual interest rate	Taux (d'intérêt annuel effectif)
Equivalent martingale measure	Mesure martingale equivalente
Filtration	Filtration
Forward (or Future)	Contrat à terme
Hedging	Couverture
Hedging strategie	Stratégie de couverture
Hybrid financial contract	Produit hybride
Interest rate	Taux d'intérêt
Maturity	Maturité
Nominal	Nominal
Option	Option
Organized markets	Marchés organisés
Over The Counter markets	Marchés de gré à gré
Overdraft	Découvert
Numéraire	Numéraire ou unité de compte
Payoff	Gain
Periodic interest rate	Taux d'intérêt périodique

English	Français
Portfolio (self-financing)	Portefeuille (auto-finançant)
Pricing (of contingent claim)	Evaluation d'actif contingent
Put Option (European put option)	Option de vente (européenne)
Return	Rendement
Risk neutral measure	Mesure de risque neutre
Security (or financial claim)	Titre financier
Short sale	Vente à découvert
Simple interest rate	Taux d'intérêt simple
Stock (or share of a stock)	Une action
Strike	Prix à terme
Swap	"Swap" (échange)
Trade discount	Escompte commercial
Underlying	Sous-jacent
Wealth process	Processus de richesse
Yield curve	Courbe des taux par terme
Zero-coupon bond	Obligation zéro-coupon

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